LING10 Statistics for Linguistics Winter 2026
Week 01: Introduction, Description of Data

SlideNum	Title
2	Welcome
12	Statistics and Linguistics
30	Examples of Quantitative Research in Linguistics
38	Measures of Centrality
53	Measures of Dispersion and Histograms
69	Quartiles and Boxplots
82	Types of Variables
92	How to describe a variable

Welcome to Stats for Ling

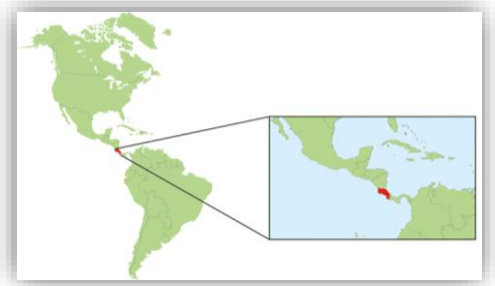
Rolando Coto-Solano. Dartmouth College

LING10, January 2026

Introductions

Name
Where are you from
Academic interests
Languages that you speak
Something fun about you

About the professor: Call me Rolando



Why was this class created?

1. Because statistics is becoming increasingly more relevant in linguistics.
2. Because you will need it if you want to become researchers.
3. Because it’s an incredible useful (and marketable) skill!

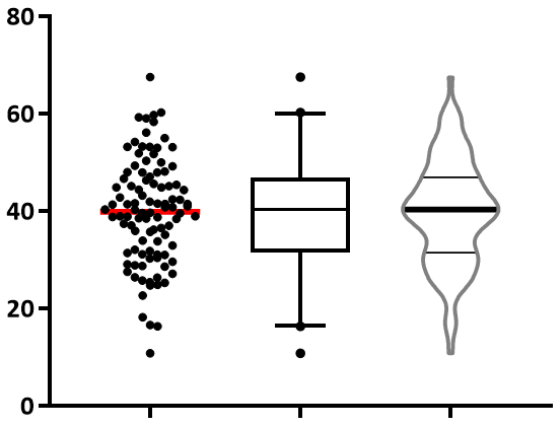
What will we study?

How not to be fooled when presented with statistics
How to interpret graphs and data
How to construct graphs
A little background math (whenever it’s needed)

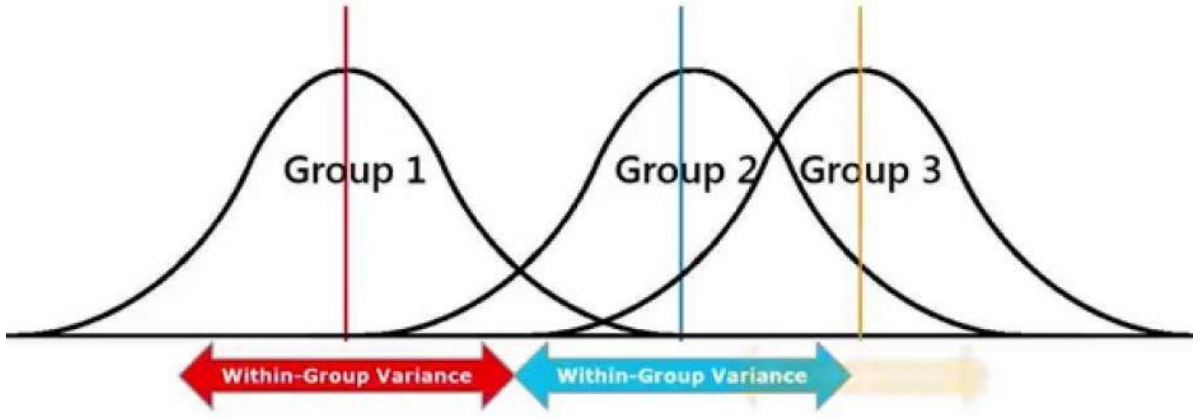
How to test whether there is a relationship between variables
e.g. Is your amount of savings related to what language you speak?

Different statistical tests you can use
and when each one is appropriate

Topics



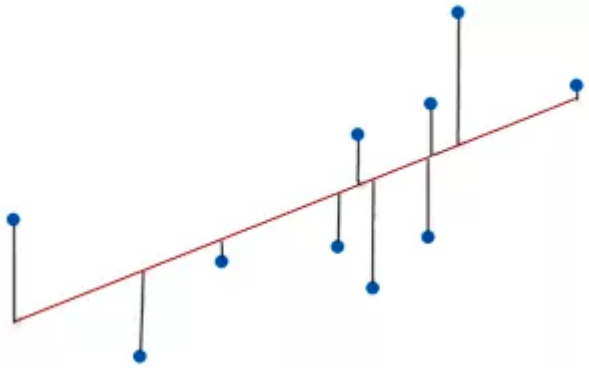
Weeks 1-2: Describe data and distributions



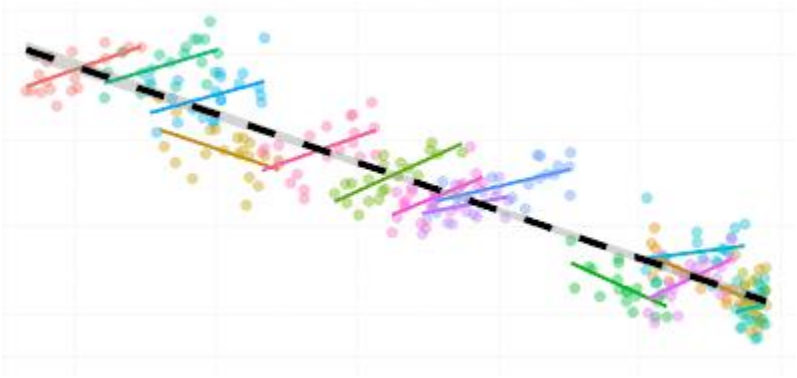
Weeks 3-5: Basic tests (t-test, Chi-square, ANOVA)

GraphPad, Learn Six Sigma

Topics



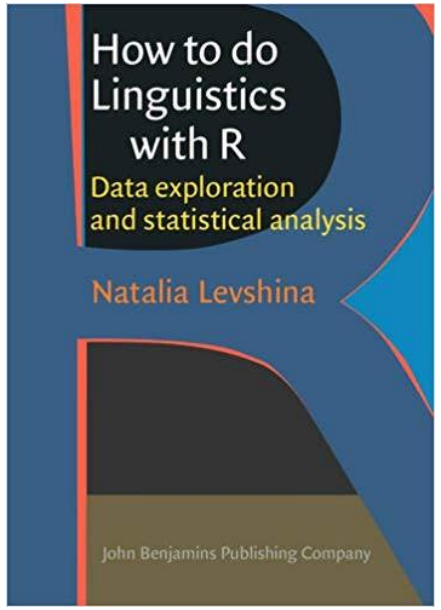
Weeks 6-8: Linear and multiple linear regression

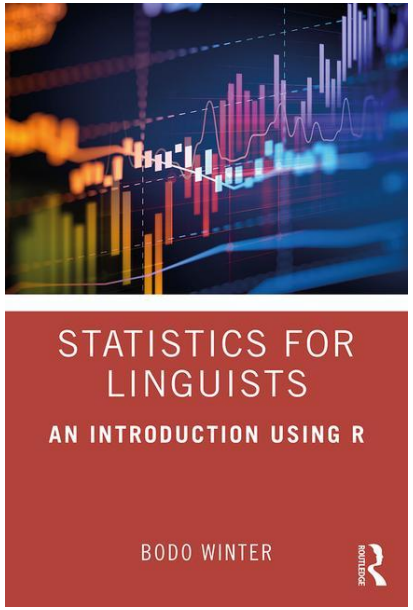


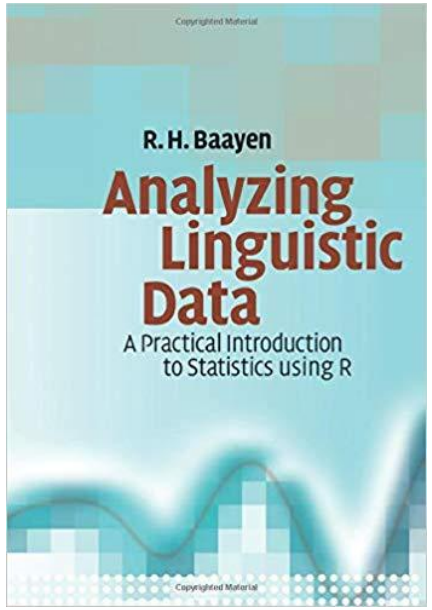
Week 9: Mixed effects and logistic linear regression

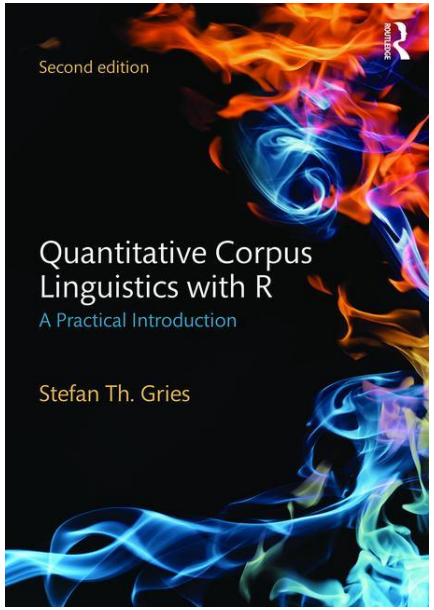
Textbook

http://www.sfs.uni-tuebingen.de/~hbaayen/publications/baayenCUPstats.pdf









R Programming

We will use the R programming language because:

- It's the *de facto* standard in scientific research.
- It's very useful in the job market.
- It leads to open and reproducible research.
- It's free and well maintained.

Please install R and RStudio for the X-Hour lab.

Syllabus

[Go to syllabus]

Frey et al. 2017

Statistics and Linguistics

Rolando Coto-Solano. Dartmouth College

LING10, January 2026

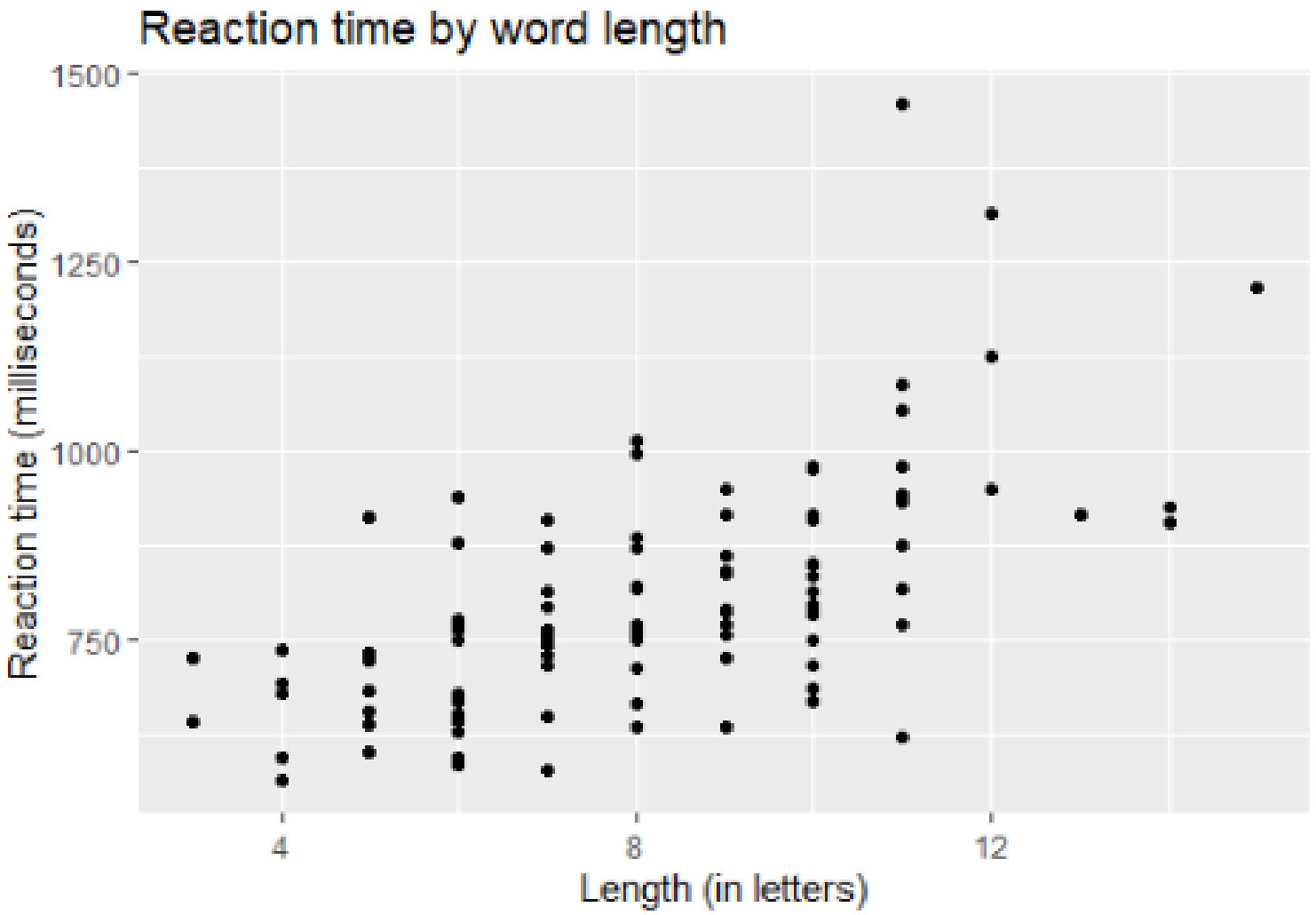
What statistics is: Are these two different?

In Columbus, OH, people sometimes say **[s]treet**,
while sometimes they say **[ʃ]treet** (Durian 2007, Johnson 2009).

An experiment was carried out where people were asked for an address. Then, the experimenter asked “where?” to elicit a second version of the word with more *emphasis*.

	Normal word	Word with emphasis
[ʃ]tr	43	14
[s]tr	77	106

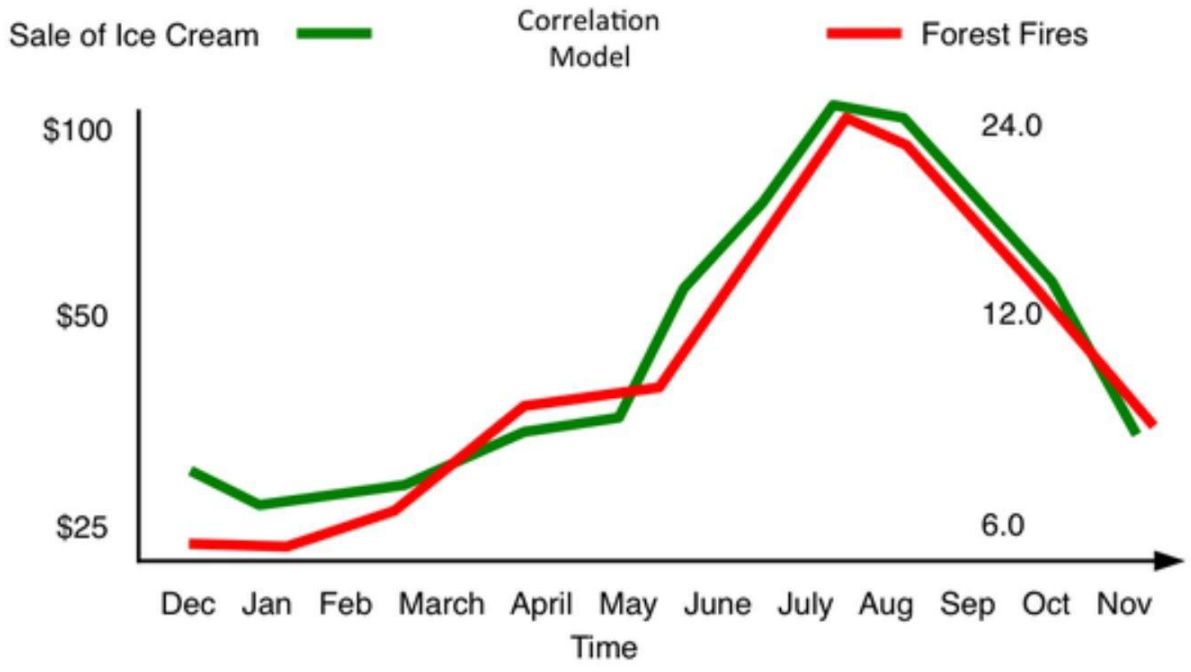
What statistics is: Are these two correlated?

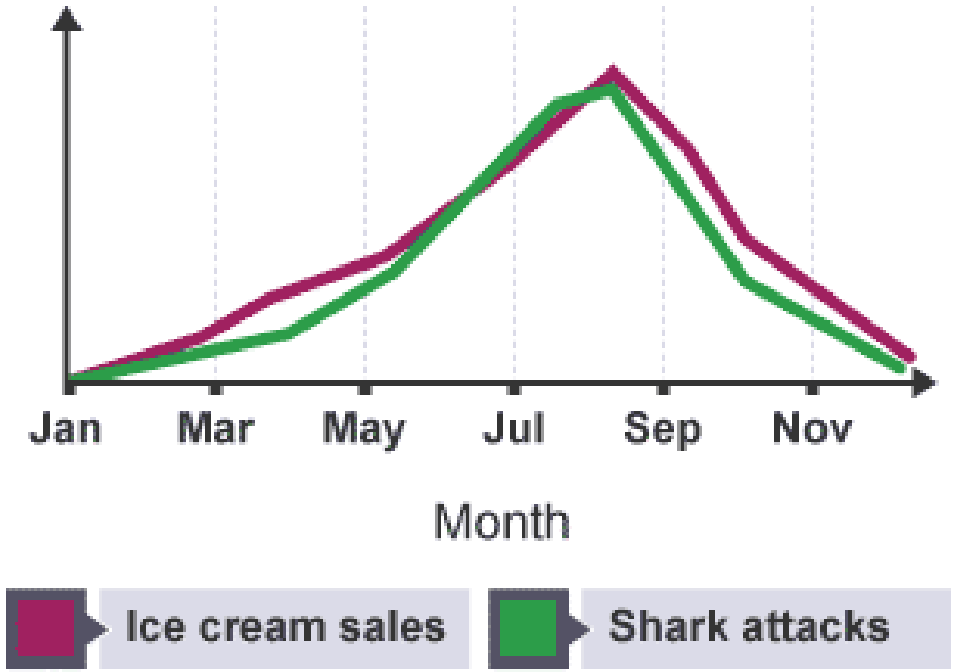


In a *lexical decision task* experiment, you sit in front of a computer and are asked if a word is a real word of English or not. We measure the **reaction time** in milliseconds.

Do longer words have slower reaction times?

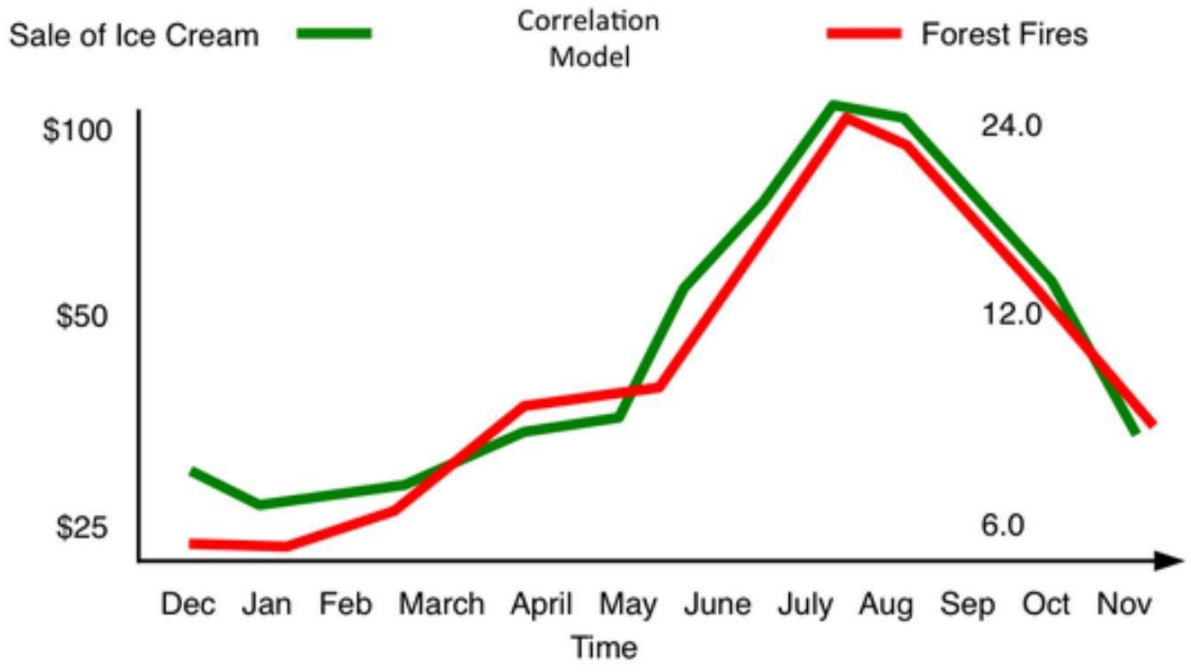
What statistics isn't: Does X cause Y?

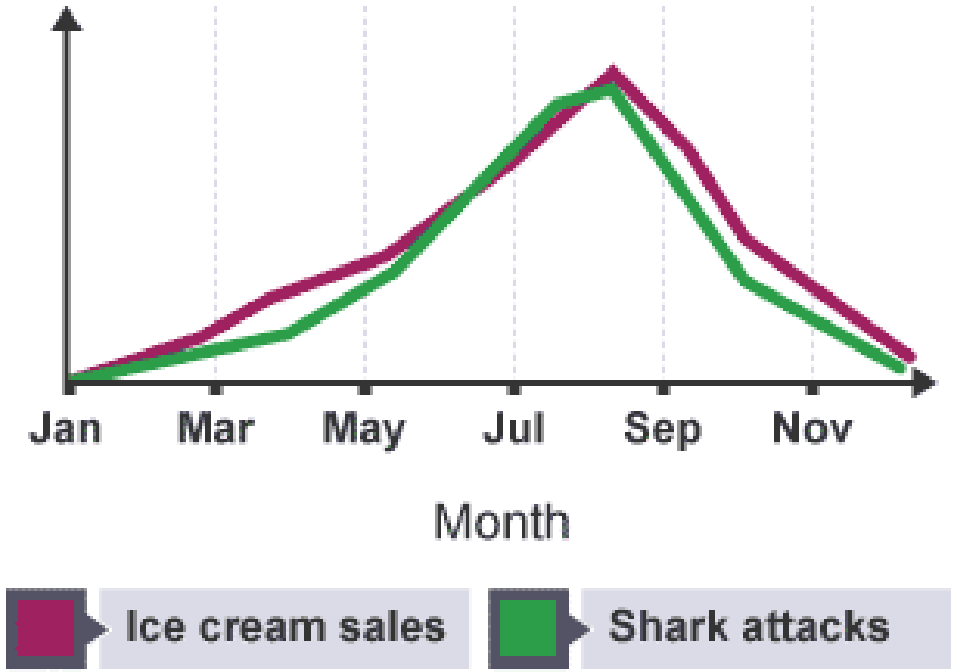




Ice cream is an extremely dangerous substance!

What statistics isn't: Does X cause Y?

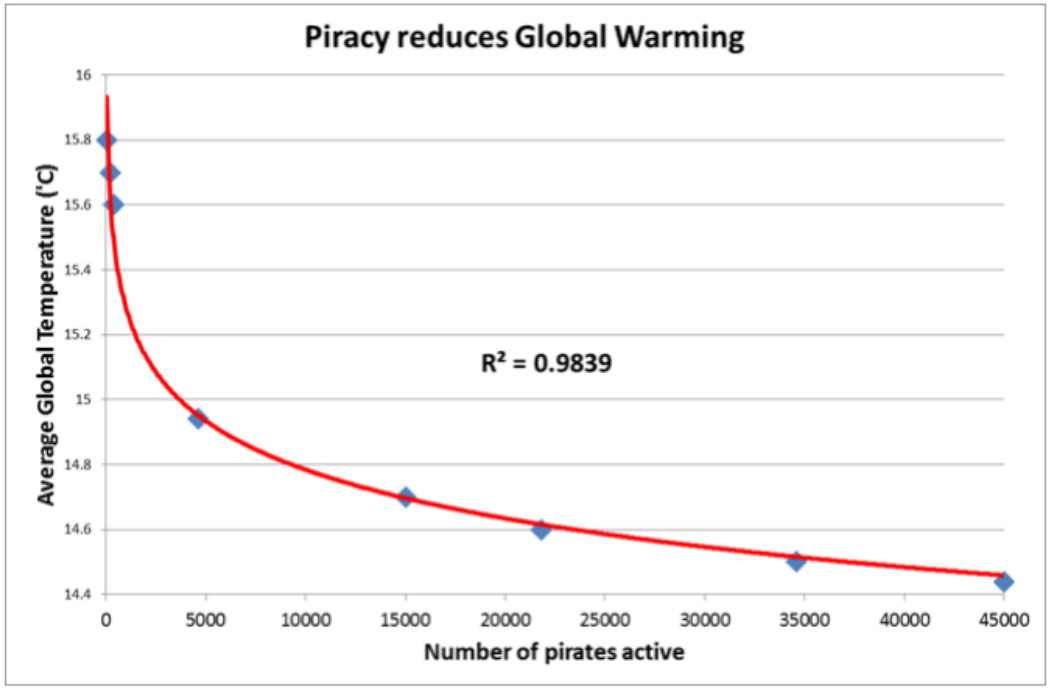


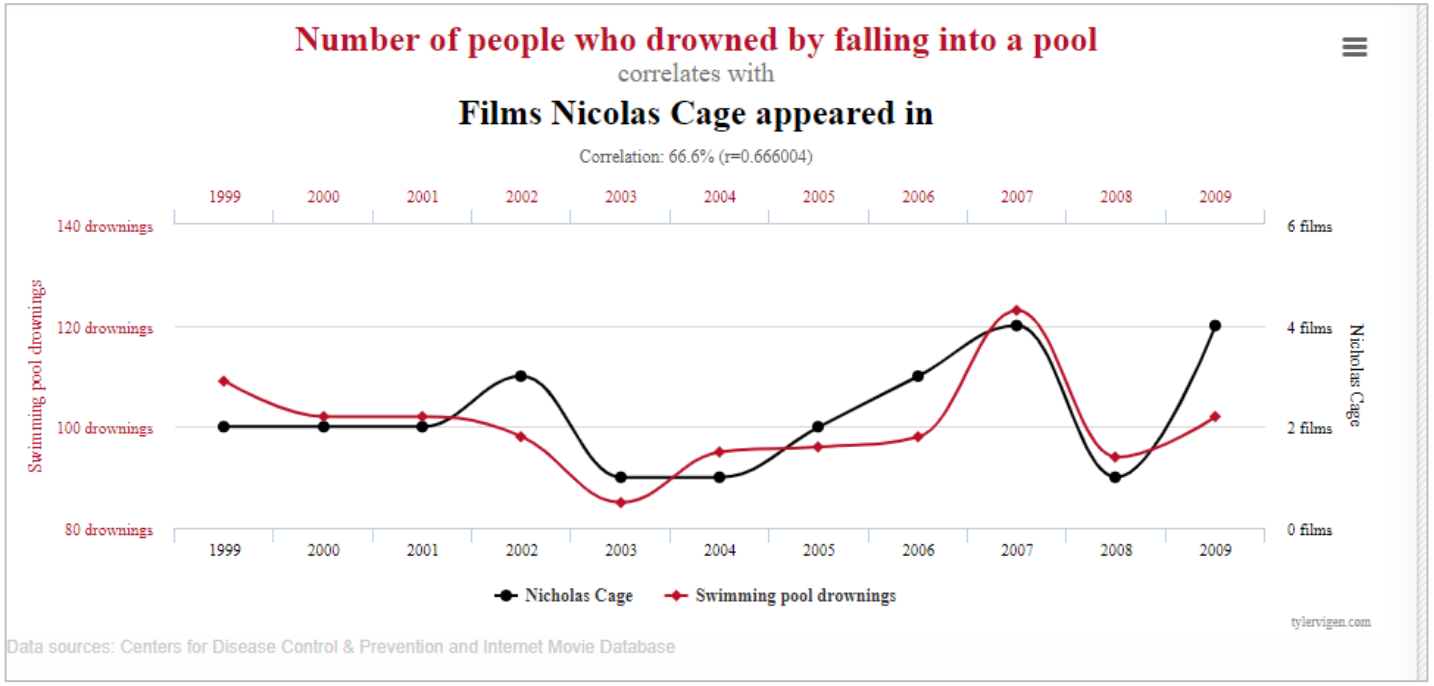


Ice cream is an extremely dangerous substance!

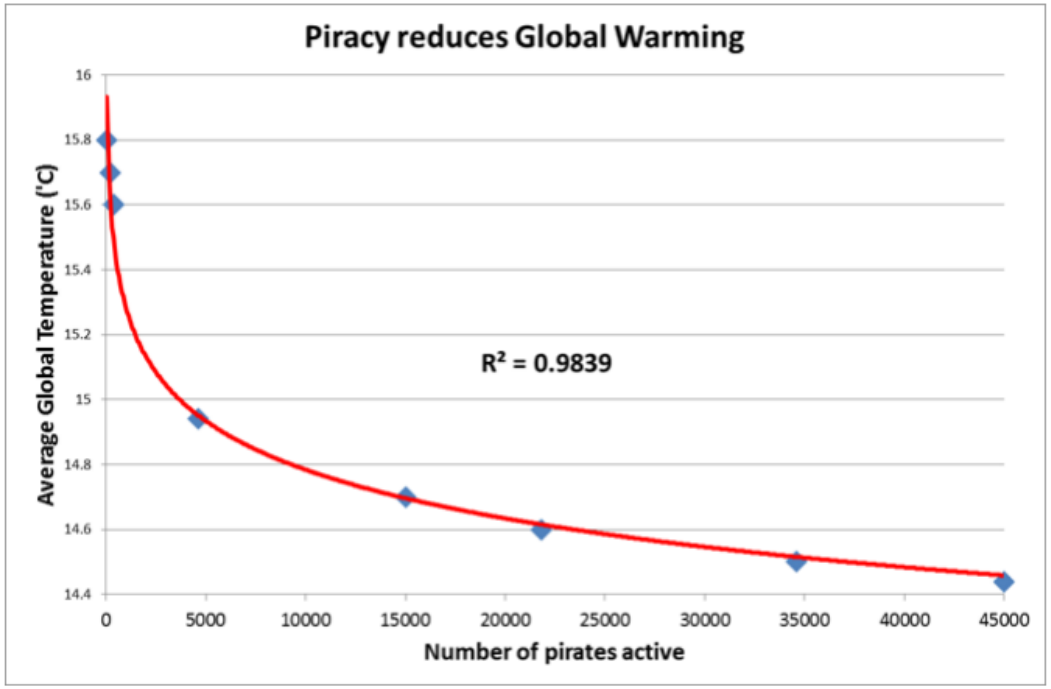
Or these two variables are **correlated to a third factor**: the weather.

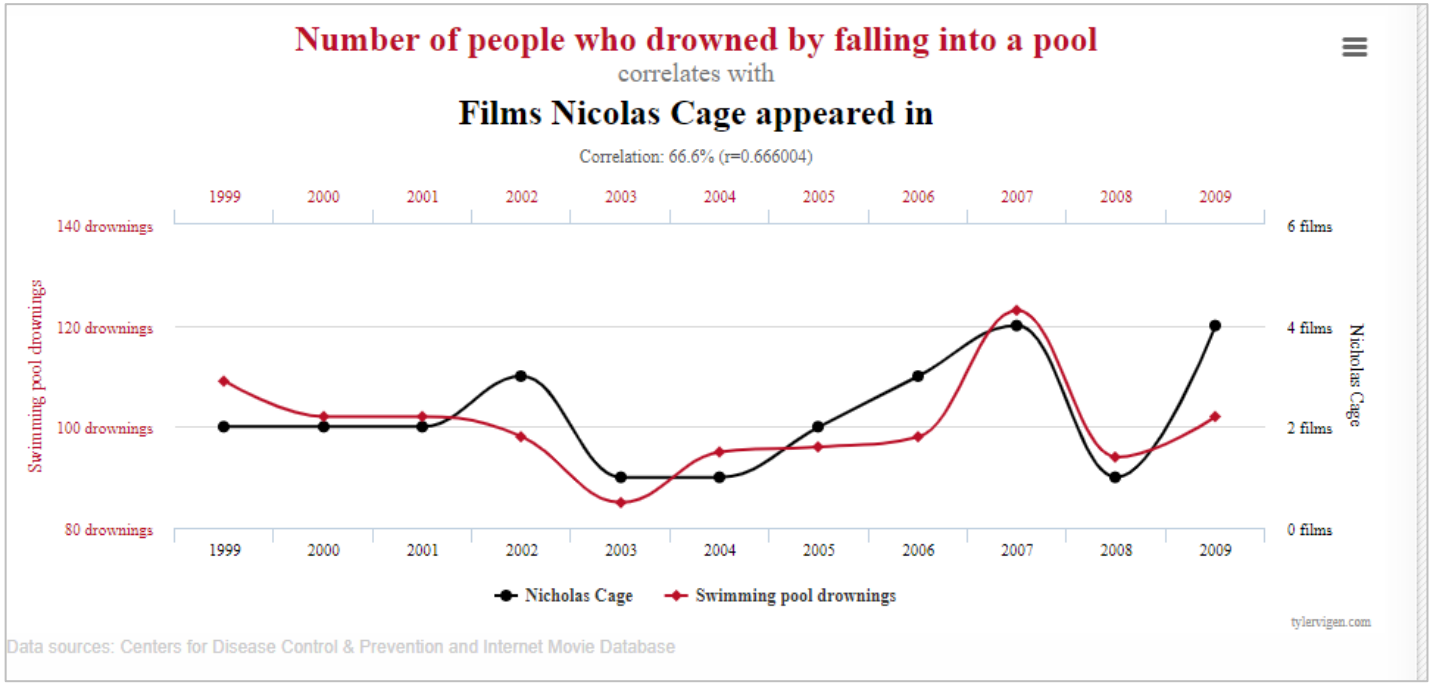
Correlation is not causation





Correlation is not causation

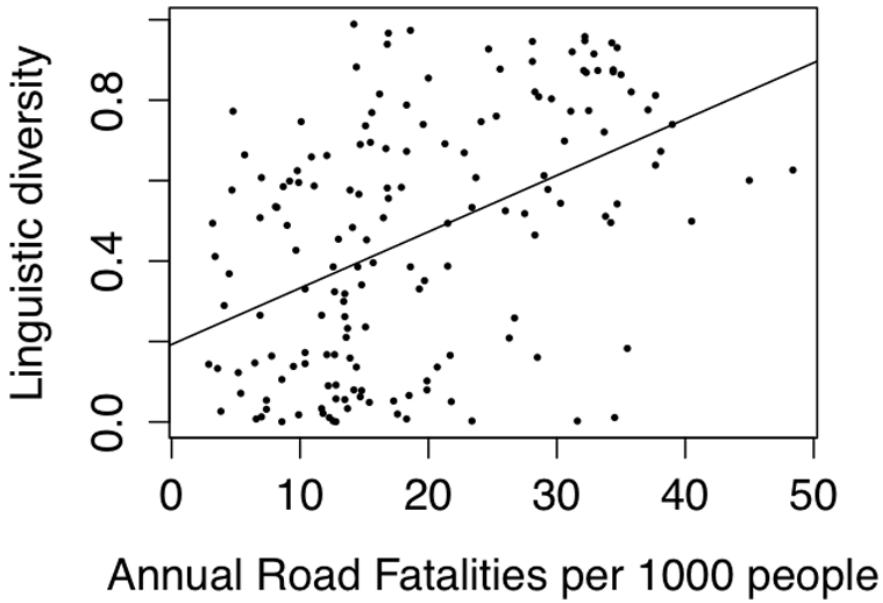


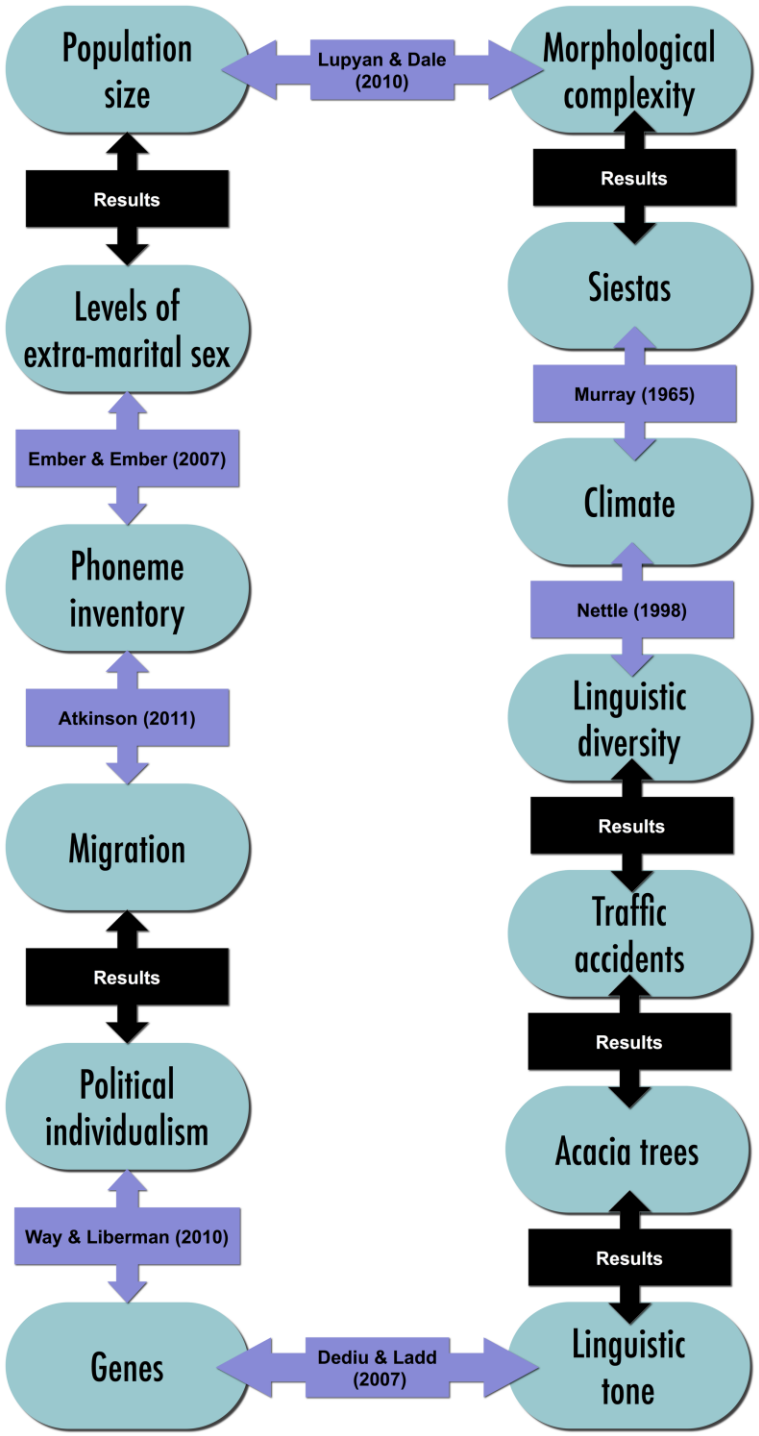


Sometimes **coincidences** just happen.
It's a big world, and accidents are
bound to happen every now and then.

https://www.tylervigen.com/spurious-correlations

Correlation is not causation





Roberts & Winters 2015

‘Forty-two!’ yelled Loonquawl. ‘Is that all you’ve got to show for seven and a half million years’ work?’

‘I checked it very thoroughly’, said the computer, ‘and that quite definitely is the answer’. ***I think the problem, to be quite honest with you, is that you’ve never actually known what the question is.’***

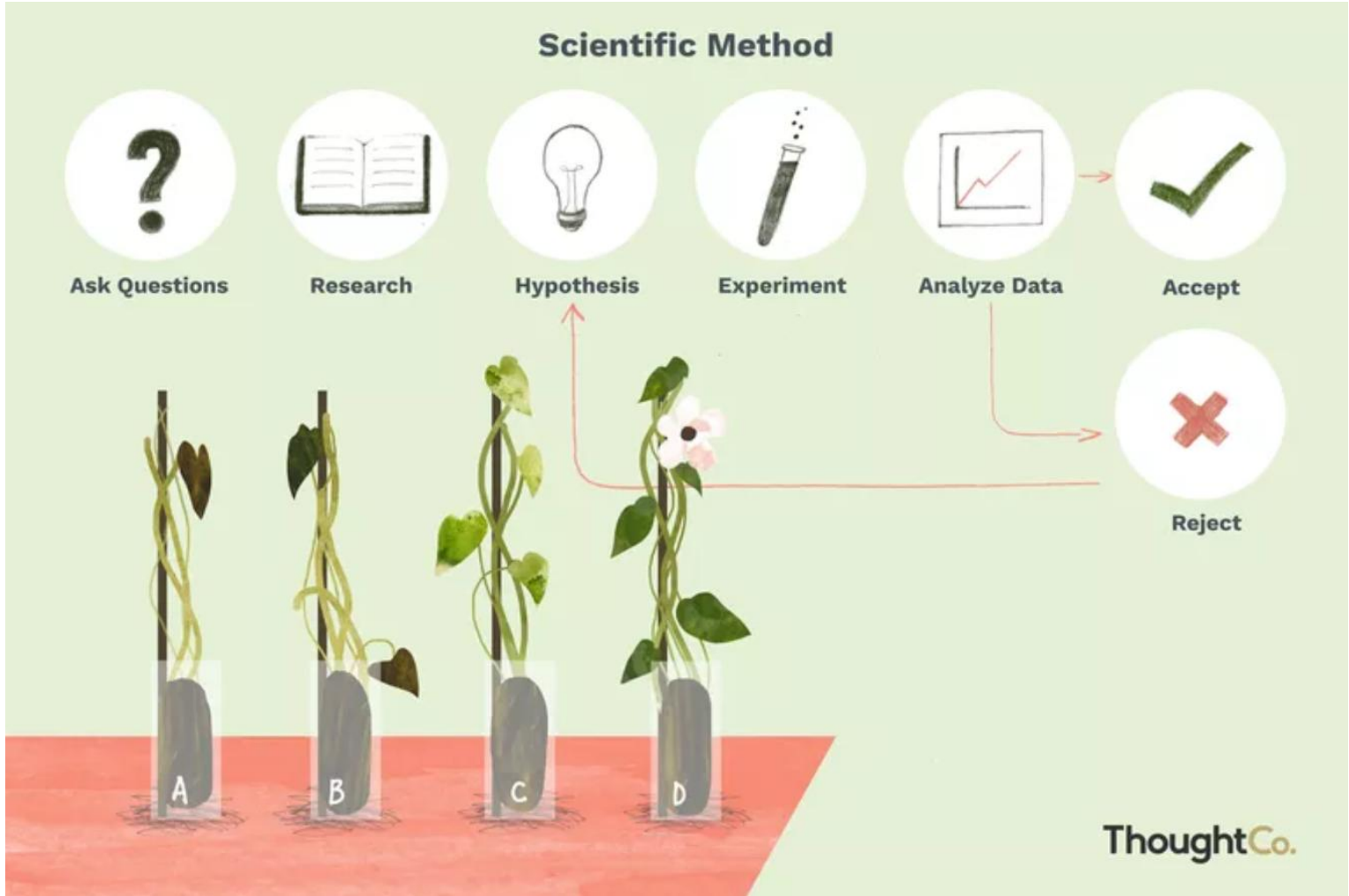
Douglas Adams, The Hitchhiker’s Guide to the Galaxy (1979)

Scientific Method and Statistics

Your knowledge of the theory of your field

Stats happens here

Your knowledge of the theory of your field, again

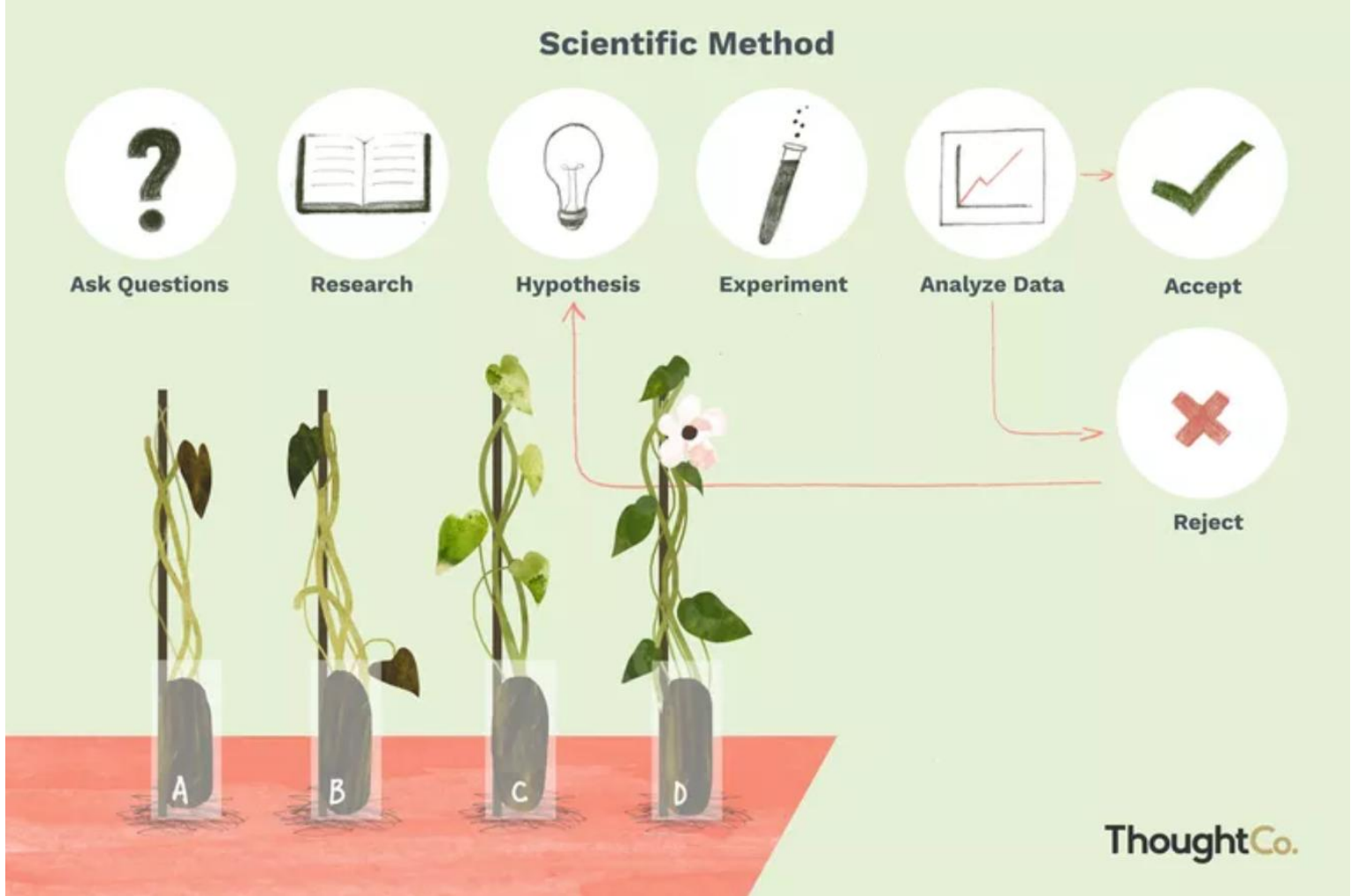


Scientific Method and Statistics

Your knowledge of the theory of your field

Stats happens here

Your knowledge of the theory of your field, again



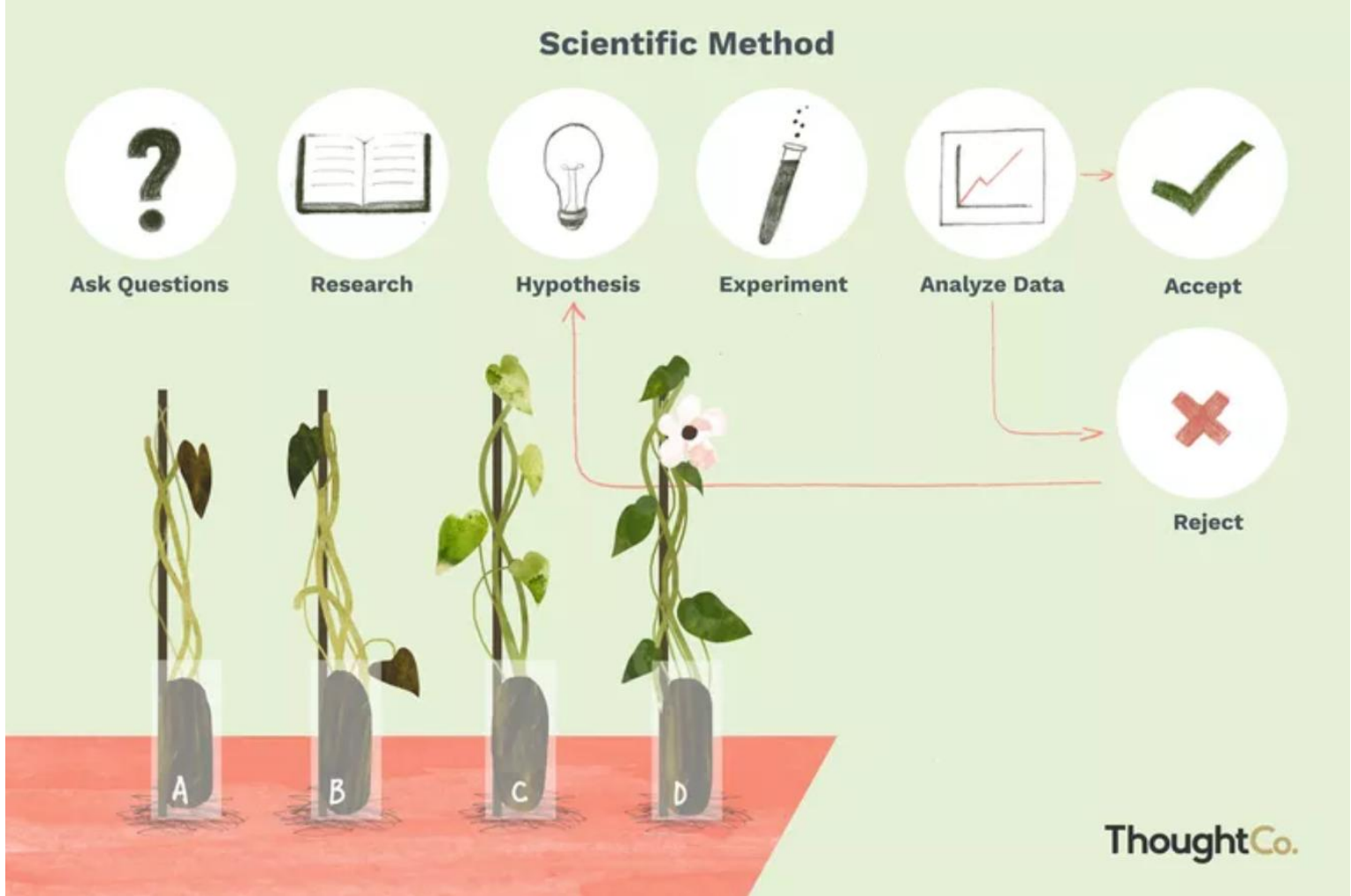
You must always start with a question, based on your expertise.

Scientific Method and Statistics

Your knowledge of the theory of your field

Stats happens here

Your knowledge of the theory of your field, again



You must always look at the data and try to interpret it before you commit to any statistical decisions.

“Why are two things correlated” is a question about your field, not about statistics.

Scientific Method and Statistics

One correlation is a good start, but we need to:

(i) Justify how this correlation fits into our theory, so we can have an actual explanation of what's happening.

(ii) We need more research, to try to disprove his correlation. If it holds, you might actually have something!

Scientific Method, Linguistics and Statistics

There are many ways to formulate questions and to provide support for our answers. (The study of what evidence is admissible for the creation of new knowledge is called **epistemology**).

All questions in linguistics are amenable to the scientific method (in that we build models of reality), but not all questions are amenable to experimentation:

Scientific Method, Linguistics and Statistics

There are many ways to formulate questions and to provide support for our answers. (The study of what evidence is admissible for the creation of new knowledge is called **epistemology**).

All questions in linguistics are amenable to the scientific method (in that we build models of reality), but not all questions are amenable to experimentation:

- Many sociolinguistics questions (e.g. ethnographies, indexicality and *persona*)
- Pure grammatical description (e.g. the verbal paradigm of a language)

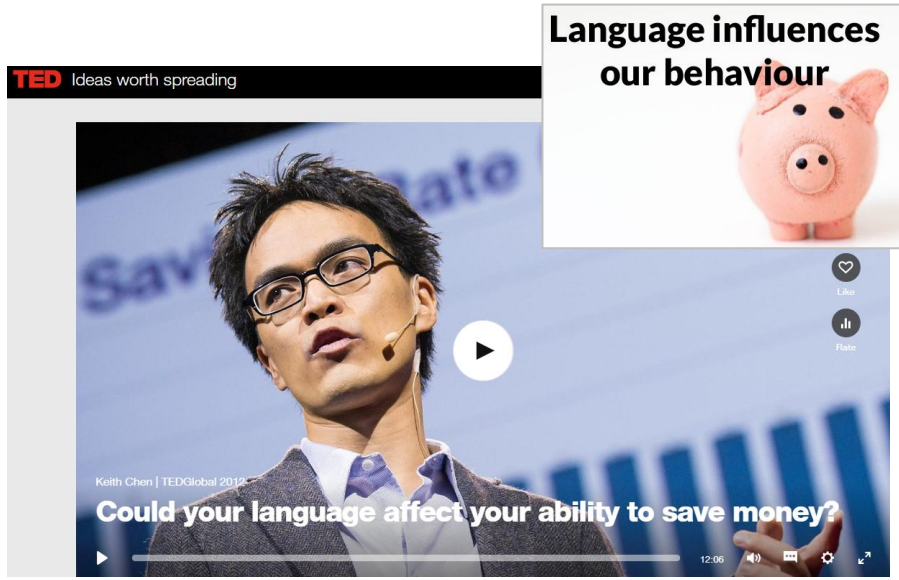
Scientific Method, Linguistics and Statistics

There are, however, many areas of linguistics where we **can** have experiments.
Which ones have you met so far?

Scientific Method, Linguistics and Statistics

Corpus linguistics

Experiments
(e.g. psycholing, phonetics)



Natural experiments

Scientific Method, Linguistics and Statistics

Quantitative methods allow you to do the following:

(1) Separating data from potential noise: Summarize trends, capture the common aspects of a series of observations as opposed to the peculiarities of each

(2) Inference: Generalize from a small set of observations to a larger universe of possible observations.

(3) Discover relationships: Find patterns in the data that can be evidence in favor or against your theory.

(4) Help you think more critically and not be duped by bad (misleading) statistics.

Quantitative Methods and Linguistics

The class is going to focus in two aspects of quantitative linguistics:

Description

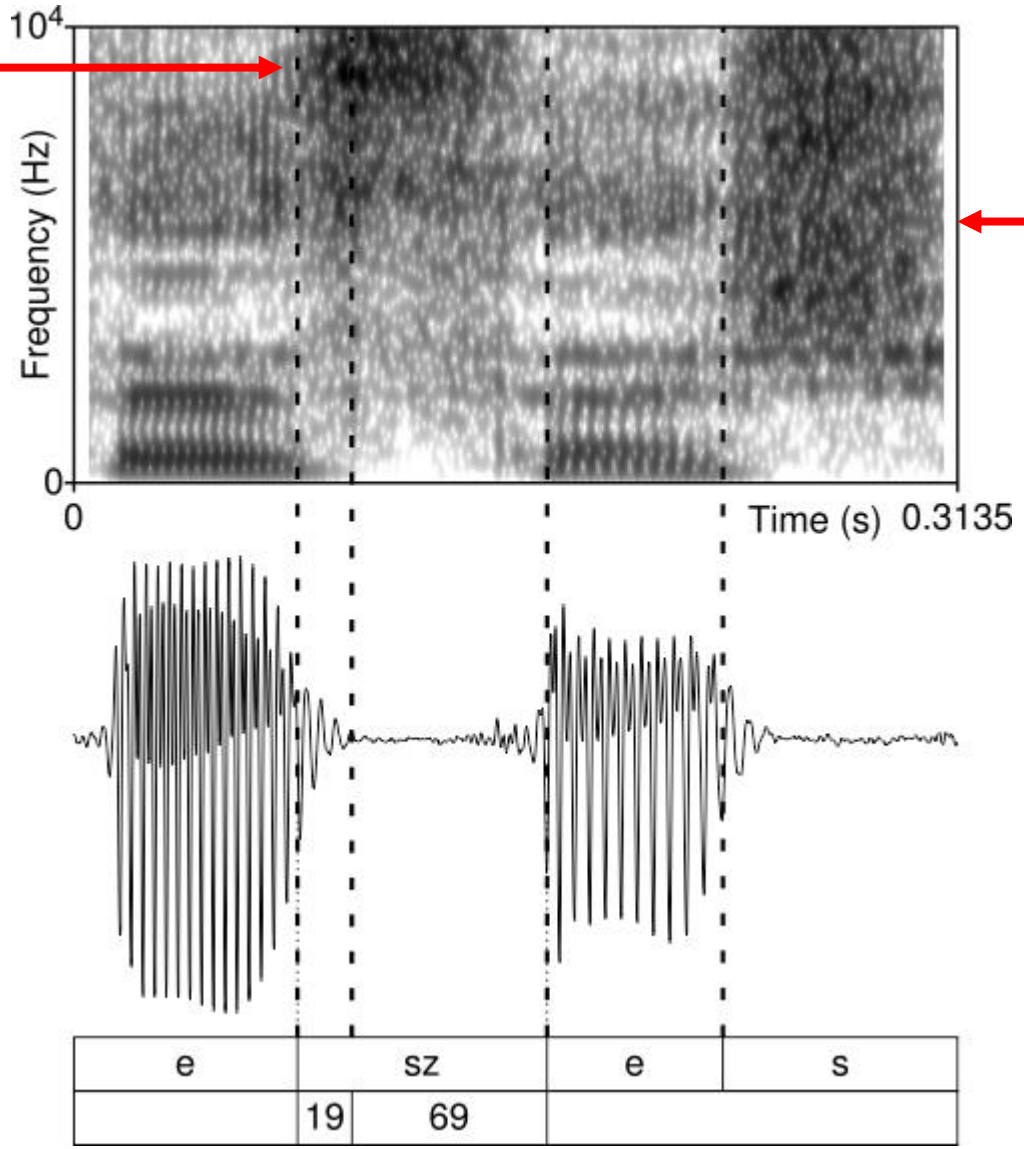
Inference

Example 1, Description: Center of Gravity of Fricatives

https://www.youtube.com/watch?v=4r7wHMg5Yjg

Example 1, Description: Center of Gravity of Fricatives

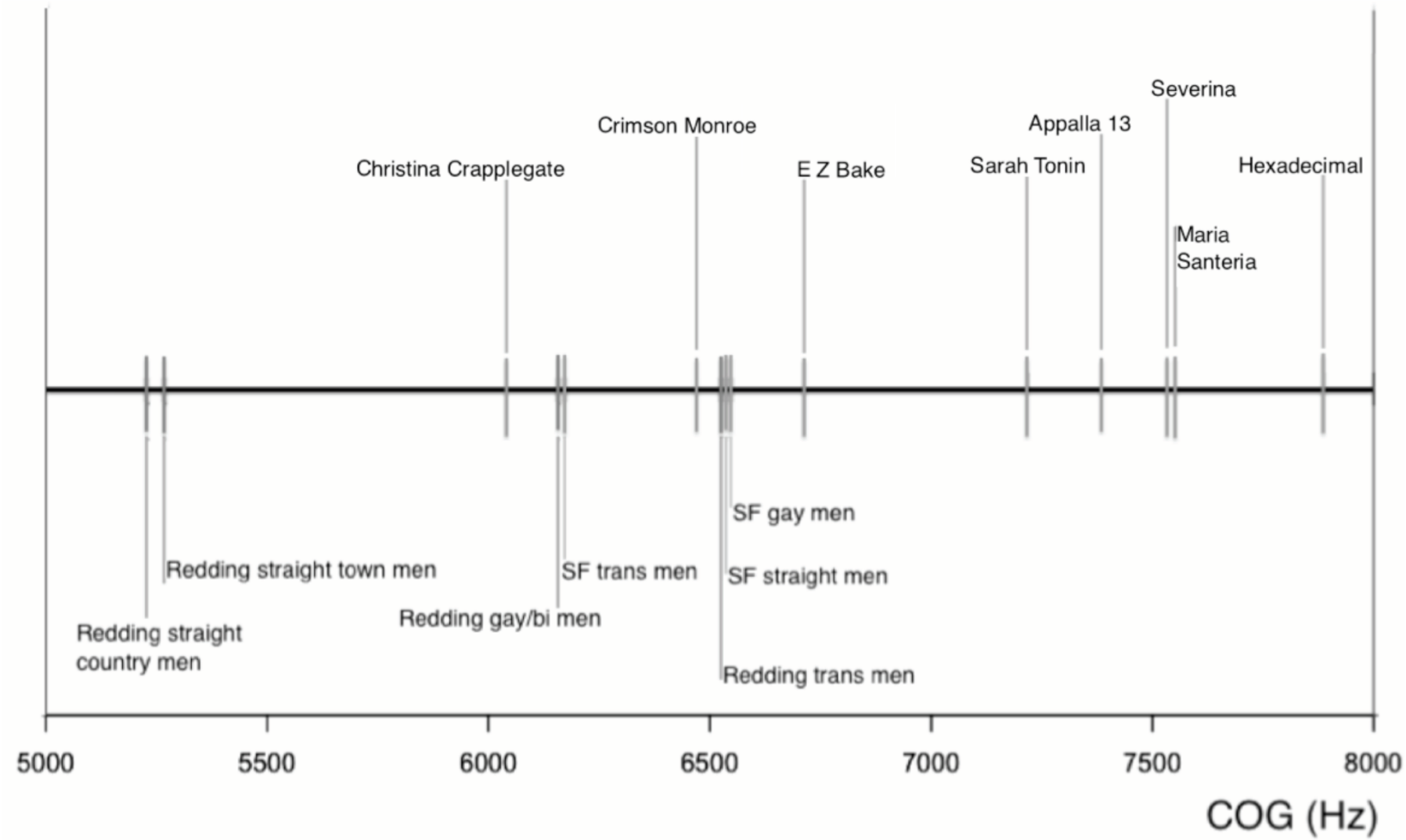
Center of Gravity of /s/



Center of Gravity of /ʃ/

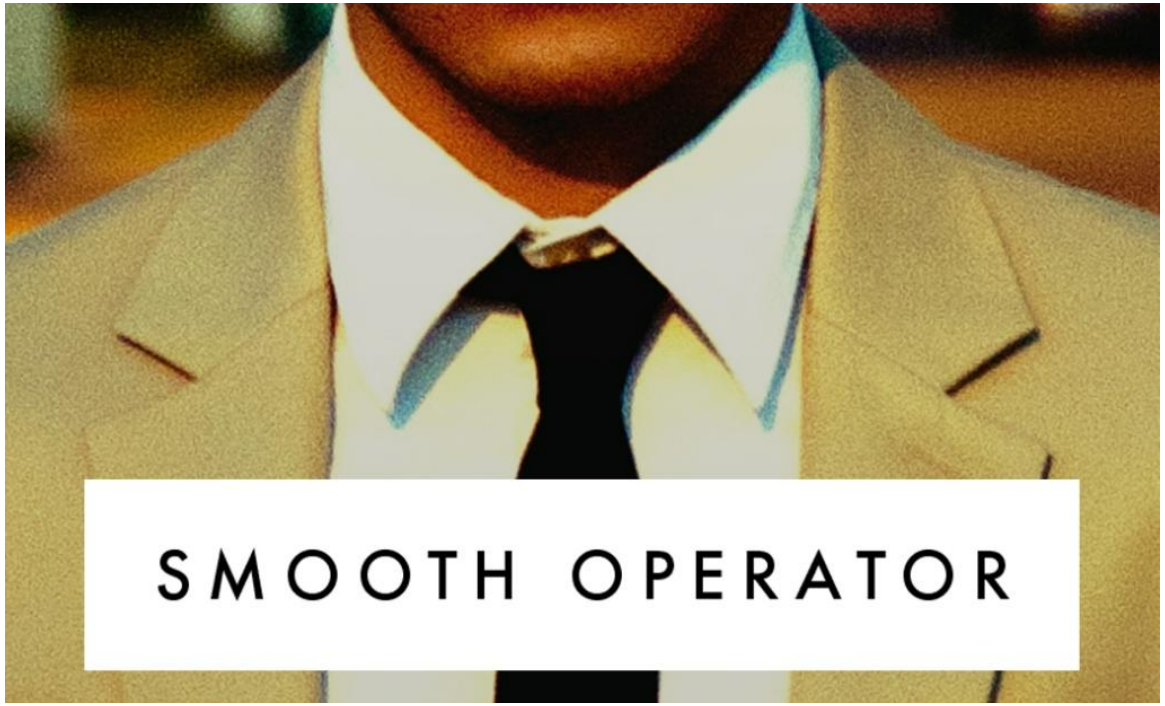
Hungarian (sz)eszes ‘alcoholic (drink)’
From Kiss (2013)

Example 1, Description: Center of Gravity of Fricatives



Data from Calder (2018), Zimman (2013) and Podesva & Van Hofwegen (2016)

Example 2, Inference: Smooth Operator in Mandarin



Zhang (2005) *A Chinese yuppie in Beijing: Phonological variation and the construction of a new professional identity*

Example 2, Inference: Smooth Operator in Mandarin

Standard versus Beijing Mandarin:

Rhotization (“oily” quality of speech)

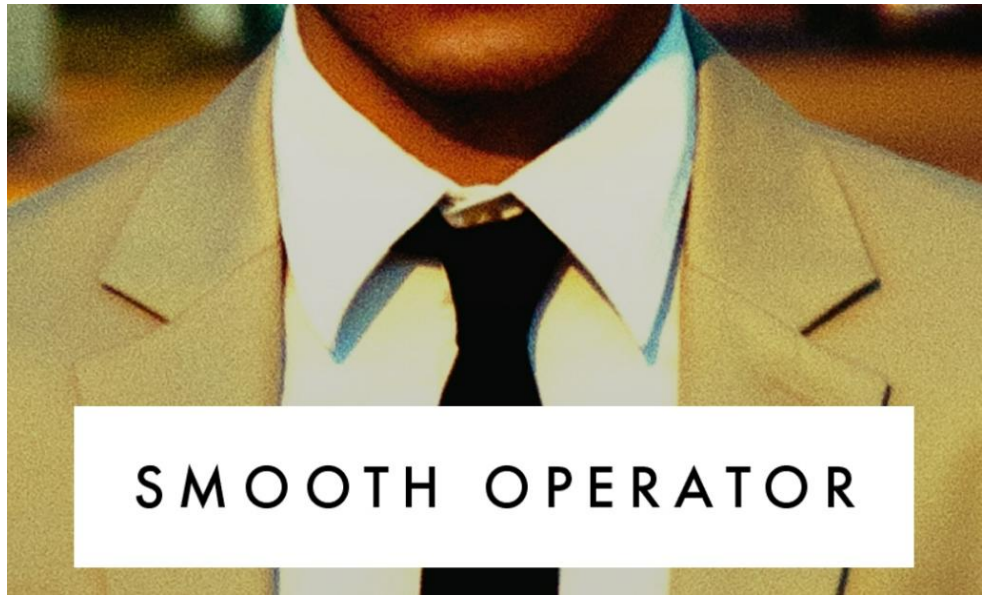
huā	*huār*	‘flower’
[hwa˥]	[hwaɹ˥]	

Interdental articulation (‘alley saunterer’)

xiànzài	*xiàn tthài*	‘now’
[ɕian˩˨tsai˩˨]	[ɕian˩˨tθai˩˨]	

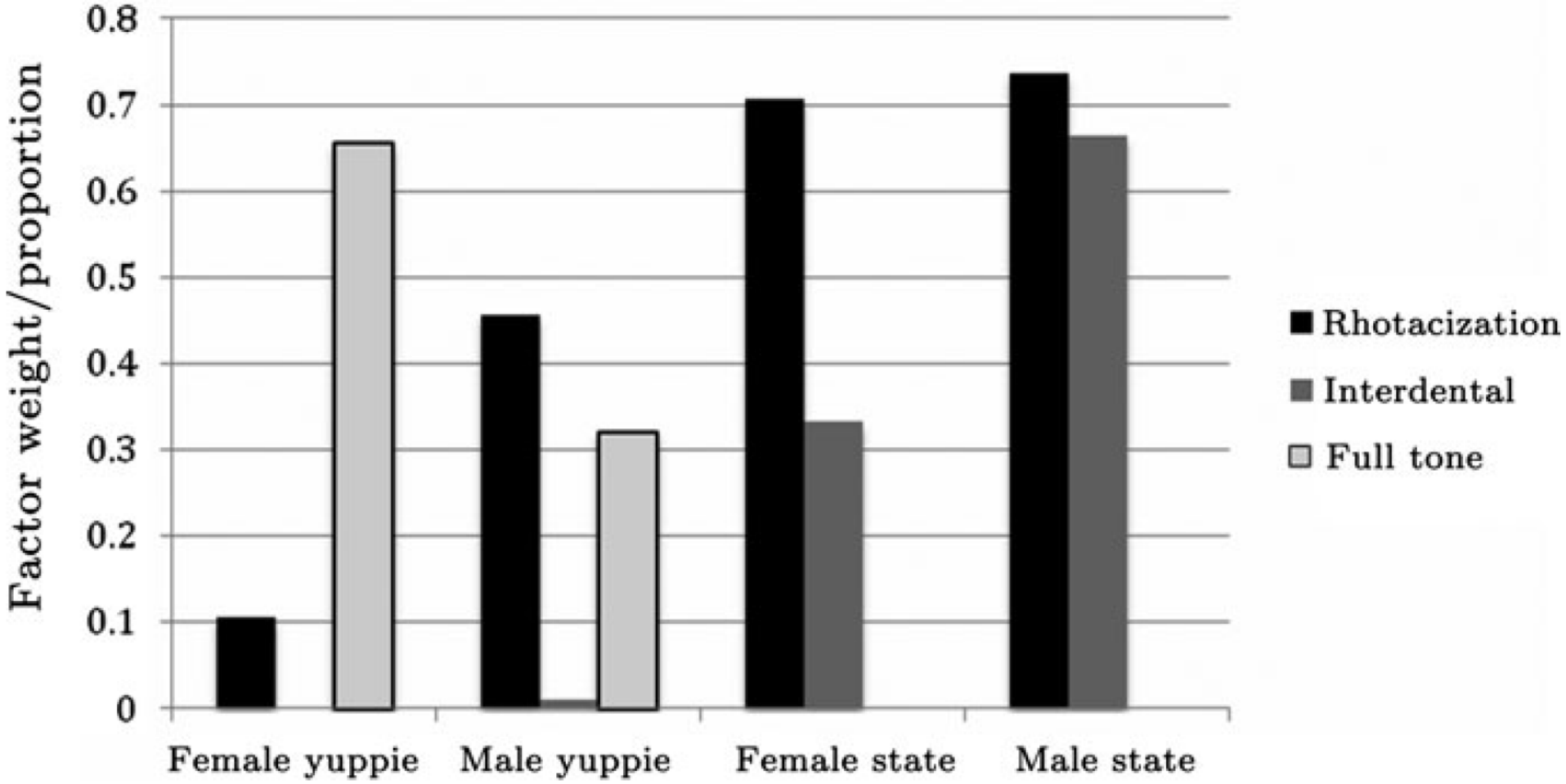
Weak tones

míngbāi	*míngbai*	‘understand’
[miŋ˧˥pai˥]	[miŋ˧˥pai˧]	



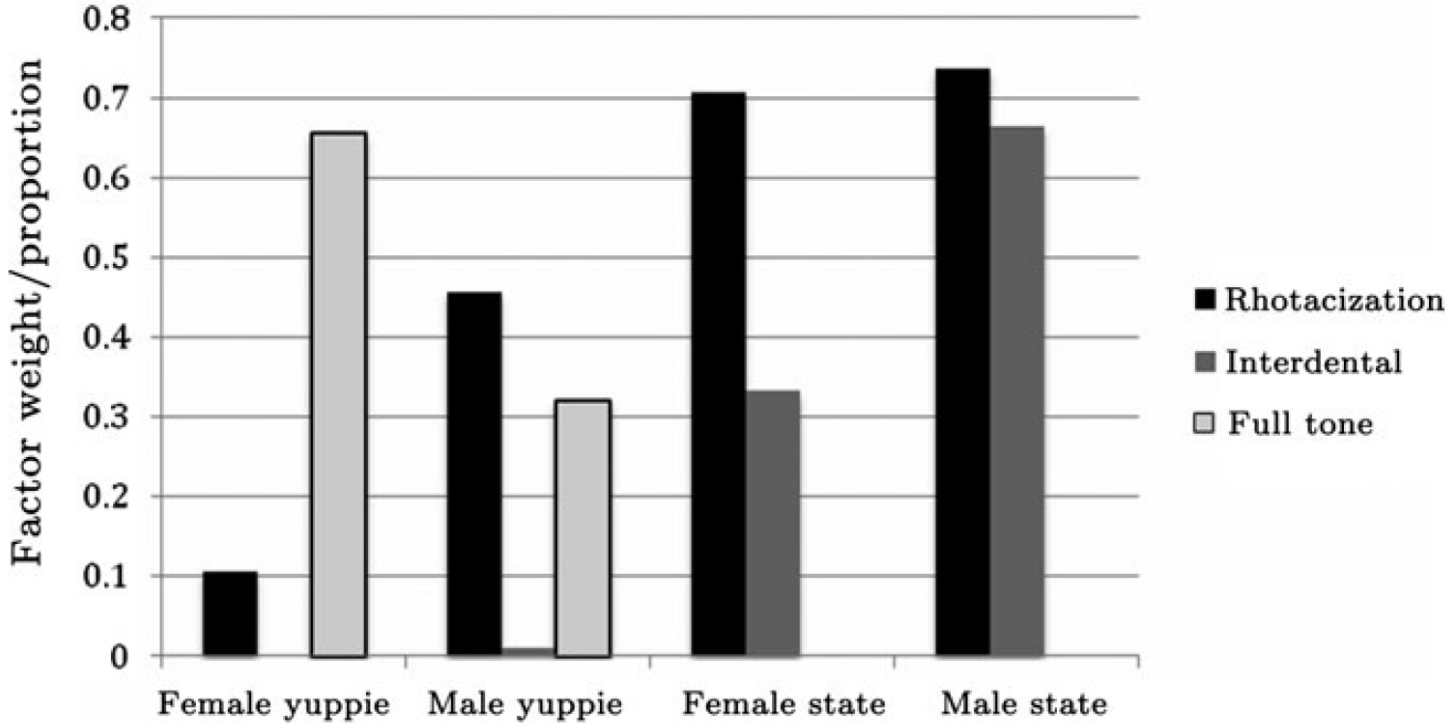
Zhang (2005) *A Chinese yuppie in Beijing: Phonological variation and the construction of a new professional identity*

Example 2, Inference: Smooth Operator in Mandarin



Zhang (2005) *A Chinese yuppie in Beijing: Phonological variation and the construction of a new professional identity*

Example 2, Inference: Smooth Operator in Mandarin



Description:

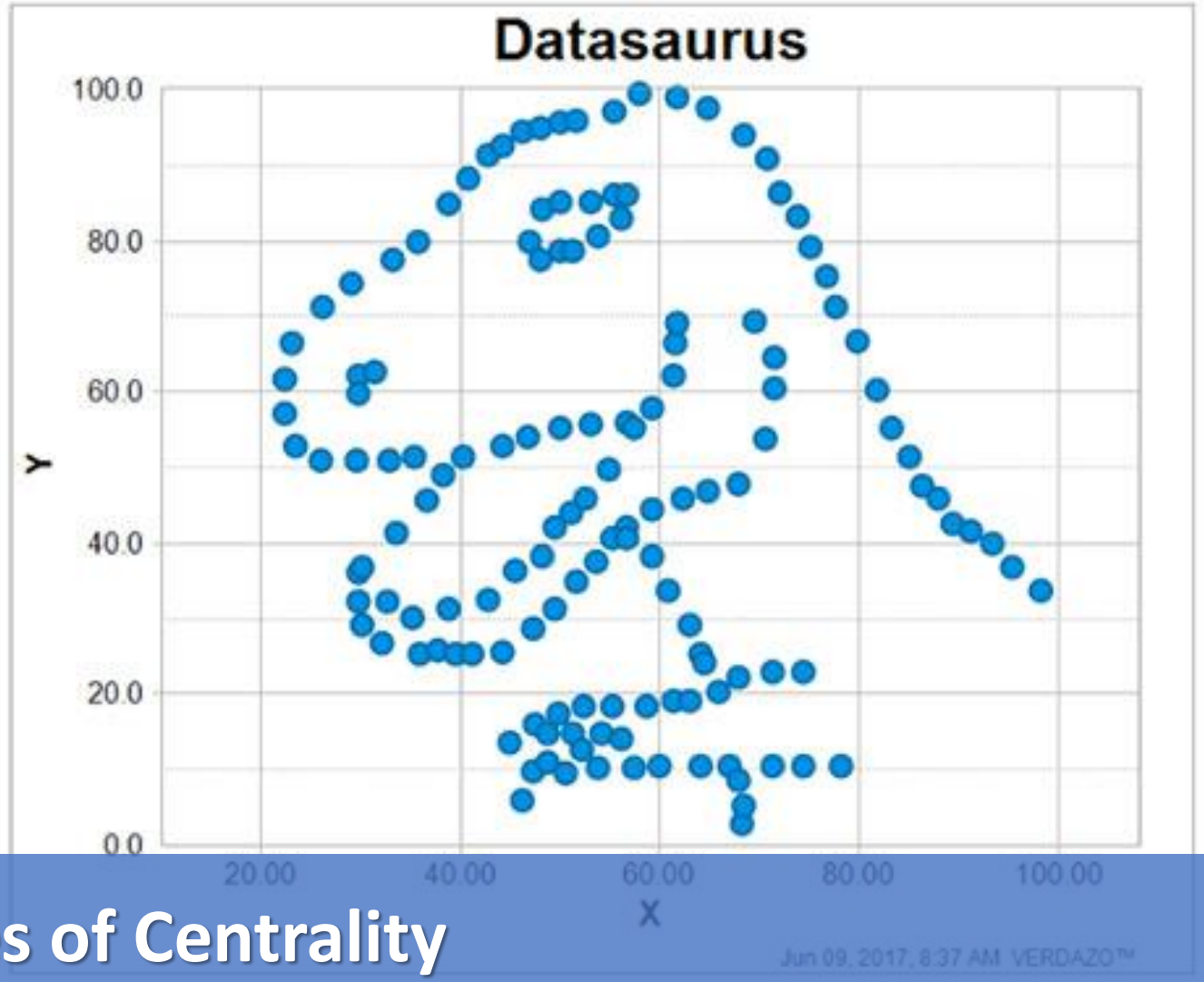
Out of the 28 people in the sample

- State employees use more of the "smooth operator" variants.
- Males have more of the "smooth operator" variants.

Inference:

- Similar participants are trending in similar directions.
- We **infer** from this that the description here will be true for the rest of the population.

Zhang (2005) *A Chinese yuppie in Beijing: Phonological variation and the construction of a new professional identity*

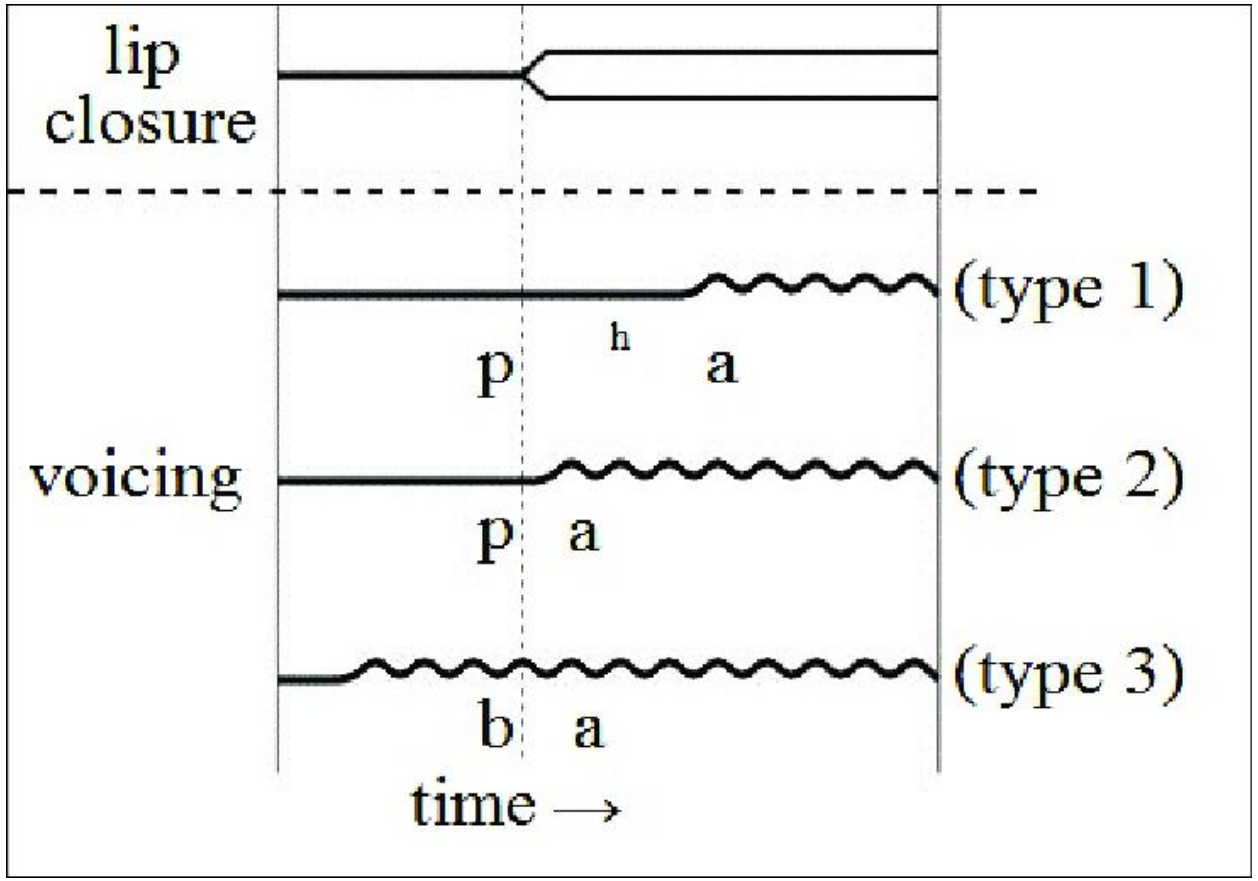


Measures of Centrality

Rolando Coto-Solano. Dartmouth College

LING10, January 2026

VOT: Voice Onset Time



Gabriel & Thiele 2017

VOT is the time between the release of a stop consonant and the start of the voicing (e.g. of the vowel).

VOT in Cherokee

VOT after /t/ in Cherokee, in milliseconds (Johnson 2009:3):

49 54 56 58 59 63 70 74 75 79 86 97 104
106 107 109 119 124 126 129 153 166 181

VOT in Cherokee

VOT after /t/ in Cherokee, in milliseconds (Johnson 2009:3):

49 54 56 58 59 63 70 74 75 79 86 97 104
106 107 109 119 124 126 129 153 166 181

Why is there **variation**?
Why isn't the measurement always the same?

VOT in Cherokee

VOT after /t/ in Cherokee, in milliseconds (Johnson 2009:3):

49 54 56 58 59 63 70 74 75 79 86 97 104
106 107 109 119 124 126 129 153 166 181

There are many factors that cause variation in our data. We need to separate the **central tendency** of the data (e.g. the mean) from the **dispersion** of the data (caused by factors that we might not understand).

VOT in Cherokee

VOT after /t/ in Cherokee, in milliseconds (Johnson 2009:3):

49 54 56 58 59 63 70 74 75 79 86 97 104
106 107 109 119 124 126 129 153 166 181

There are many factors that cause variation in our data. We need to separate the **central tendency** of the data (e.g. the mean) from the **dispersion** of the data (caused by factors that we might not understand).

Mean = 97.6

VOT in Cherokee

VOT after /t/ in Cherokee, in milliseconds (Johnson 2009:3):

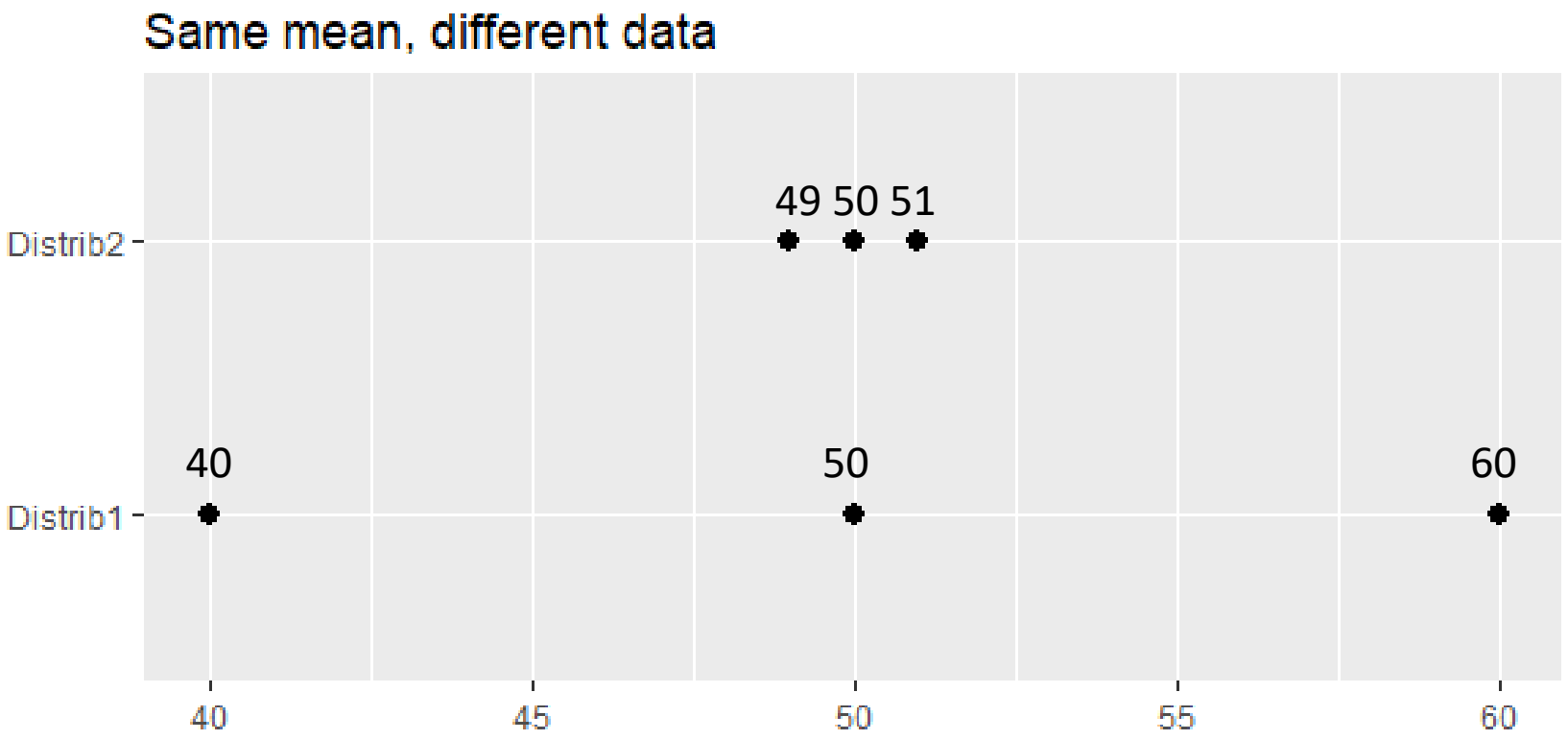
49 54 56 58 59 63 70 74 75 79 86 97 104
106 107 109 119 124 126 129 153 166 181

We gain information by shedding the details and explaining the tendency of the data in just one number.

This is very counterintuitive! We discard information (the peculiarities of each measurement), but in doing so we are also getting rid of thousands of factors we cannot control.

But the mean is not enough!

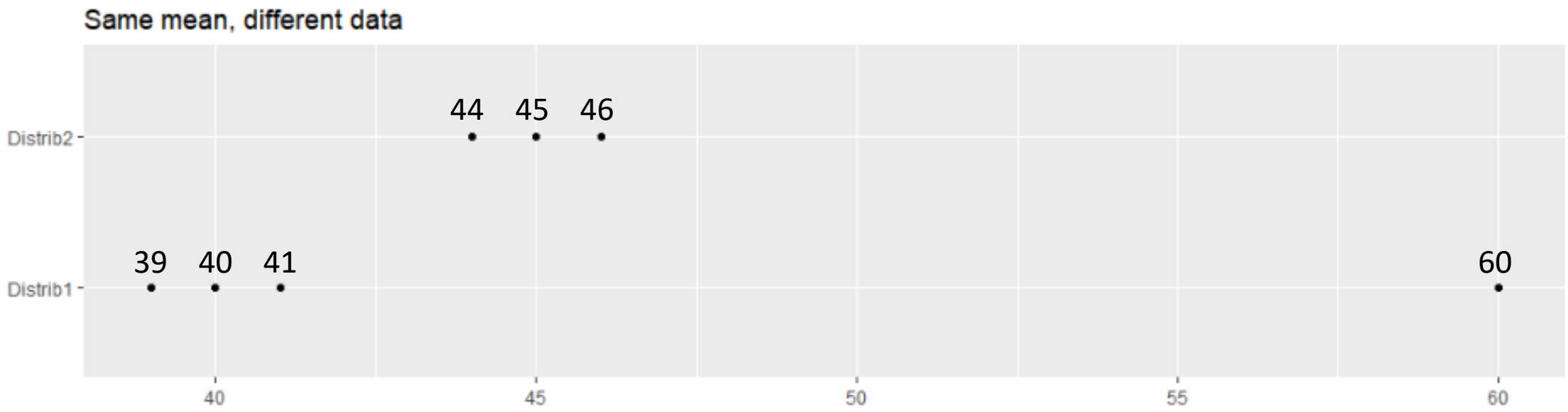
The same mean can hide different dispersion patterns in the data.



Mean = 50

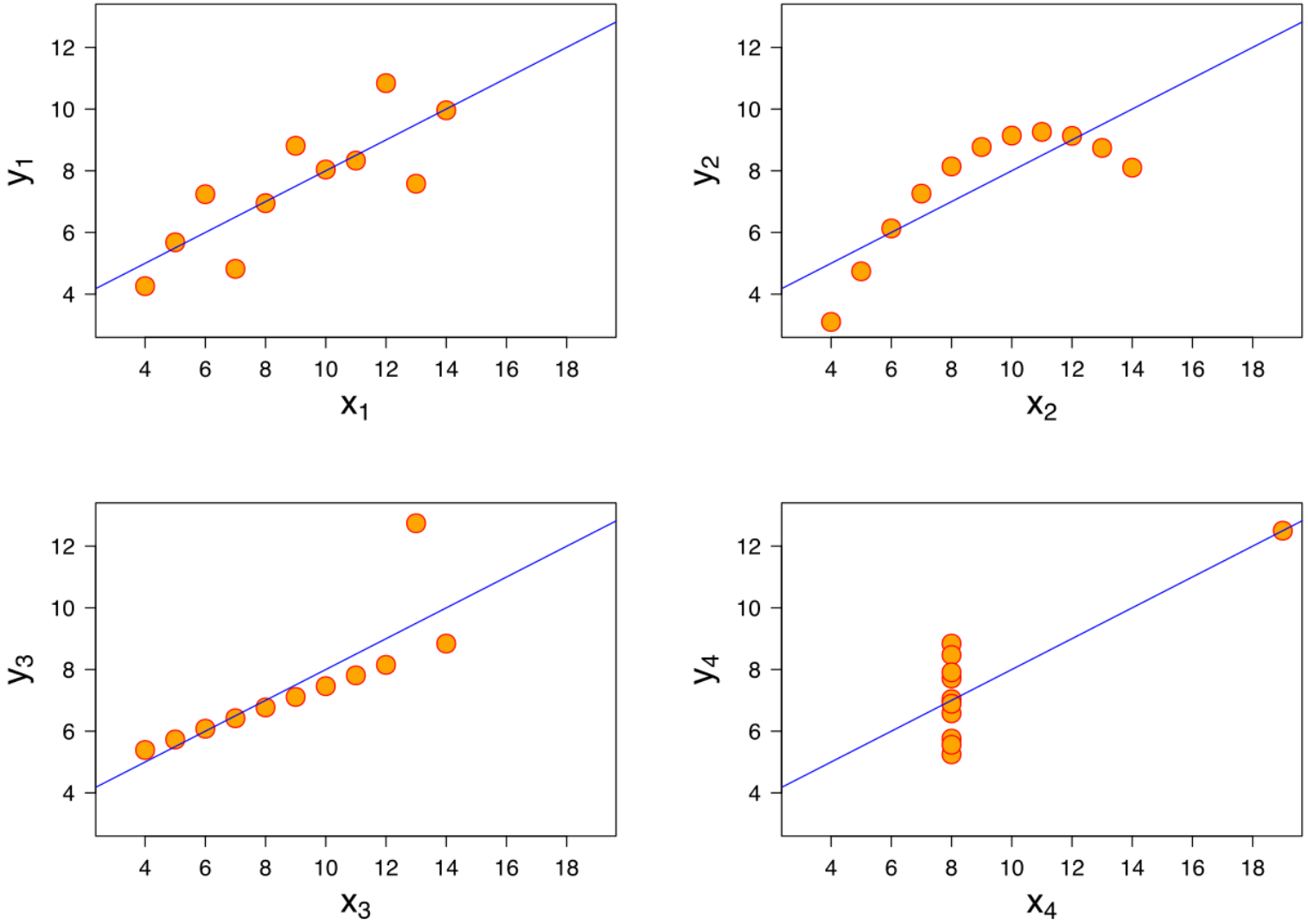
But the mean is not enough!

The same mean can hide the **skew** of the data.



Mean = 45

Anscombe's Quartet (Different data, same summary statistics)

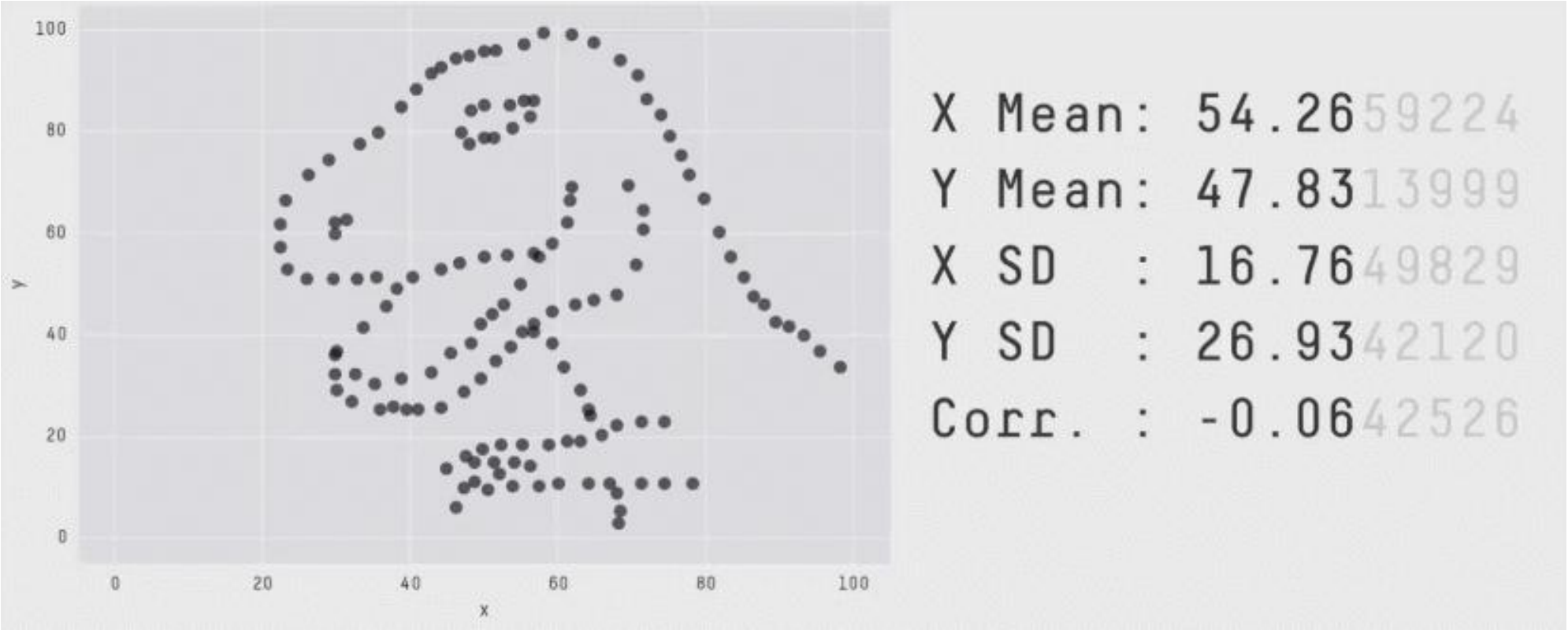


Mean of x = 9

Mean of y = 11

Schutz & Avenue; CC-BY-SA 3.0

Datasaurus! (Different data, same summary statistics)



Matejka, J., & Fitzmaurice, G. (2017, May). Same stats, different graphs: generating datasets with varied appearance and identical statistics through simulated annealing. In *Proceedings of the 2017 CHI conference on human factors in computing systems* (pp. 1290-1294).

Other measures of centrality: Median

The **median** is the “middle” value.

In the set [1,5,**9**,10,11],
the median is **9** because
it’s the 3rd out of 5 elements.

In the set [1,**5**,**10**,11],
the median is **7.5** because
the “middle” is in between the 2nd and 3rd element.

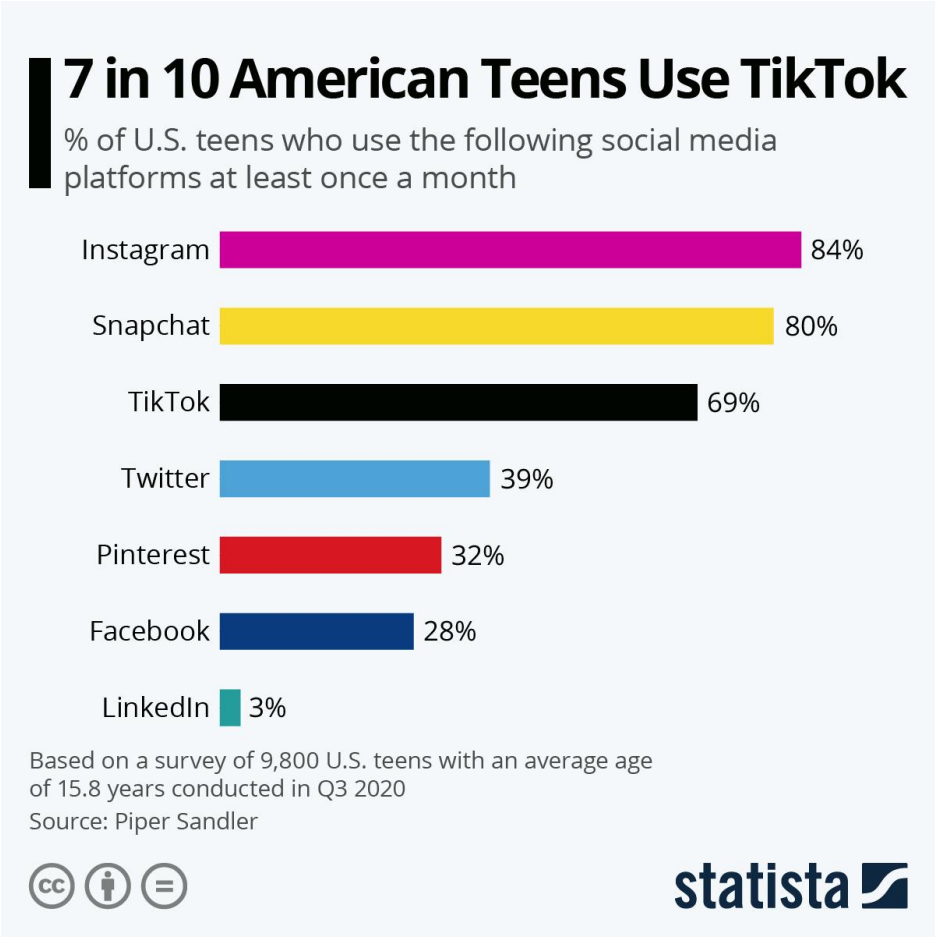
Other measures of centrality: Median

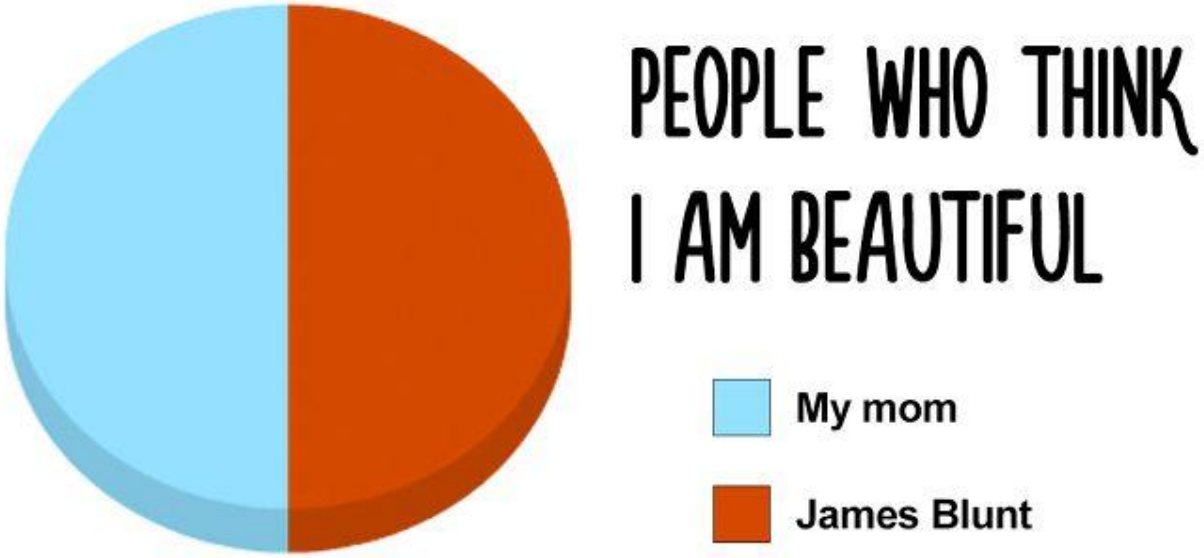
The **median** is more resilient to skewed data.

In the set:
[1,1,1,1,1,**1**,1,1,1,1,100]

Mean = 10
Median = 1

Other measures of centrality: Mode





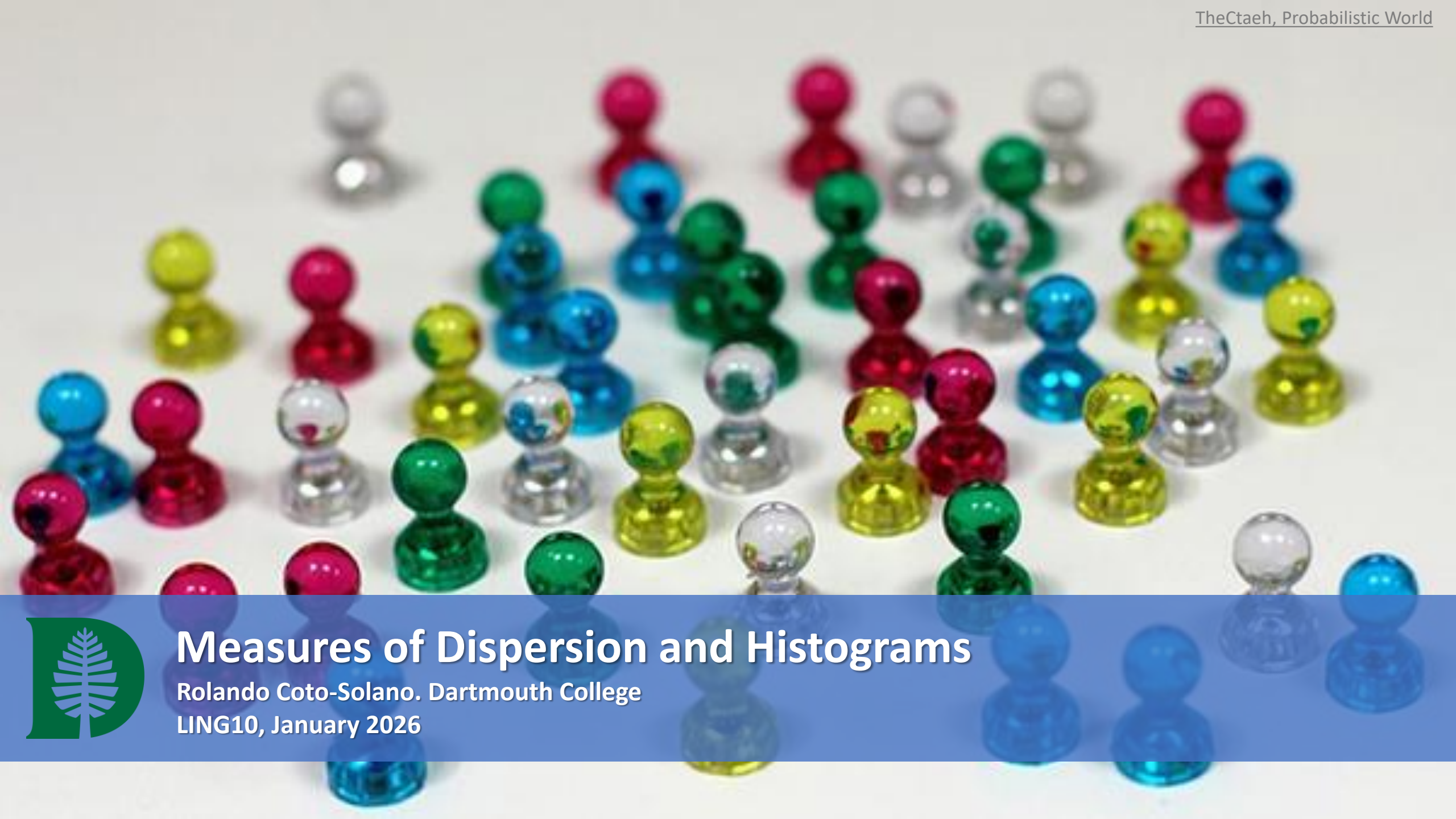
The **modal class** or **mode** is the category with the highest frequency.

Summary

We study the **central tendency** of our data so that we can eliminate unexplainable variation.

Measures of centrality:
Mean
Median
Mode

The mean might not be enough! It can hide dispersion and it can be affected by **skewed** data.



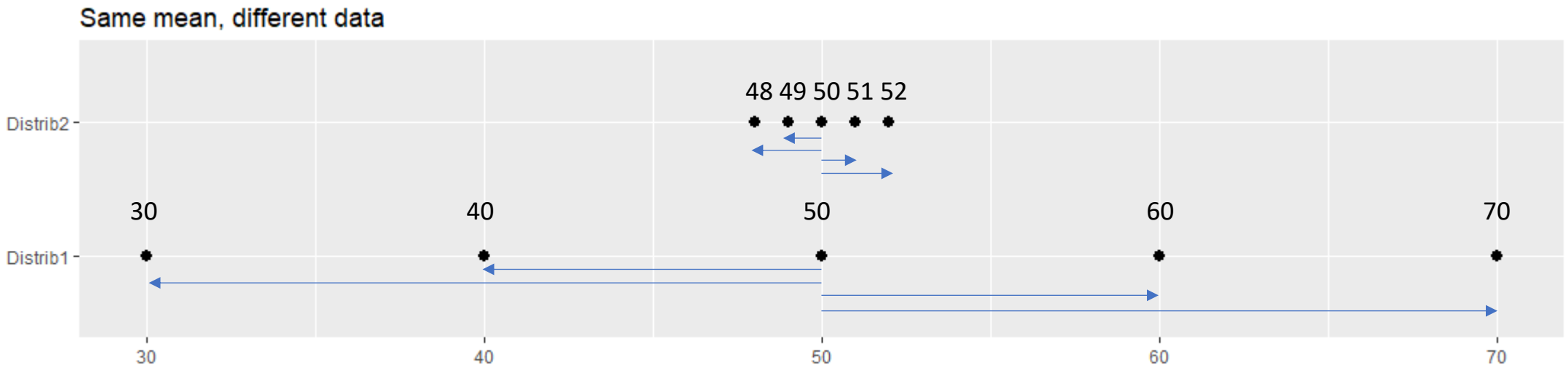
TheCtaeh, Probabilistic World

Measures of Dispersion and Histograms

Rolando Coto-Solano. Dartmouth College

LING10, January 2026

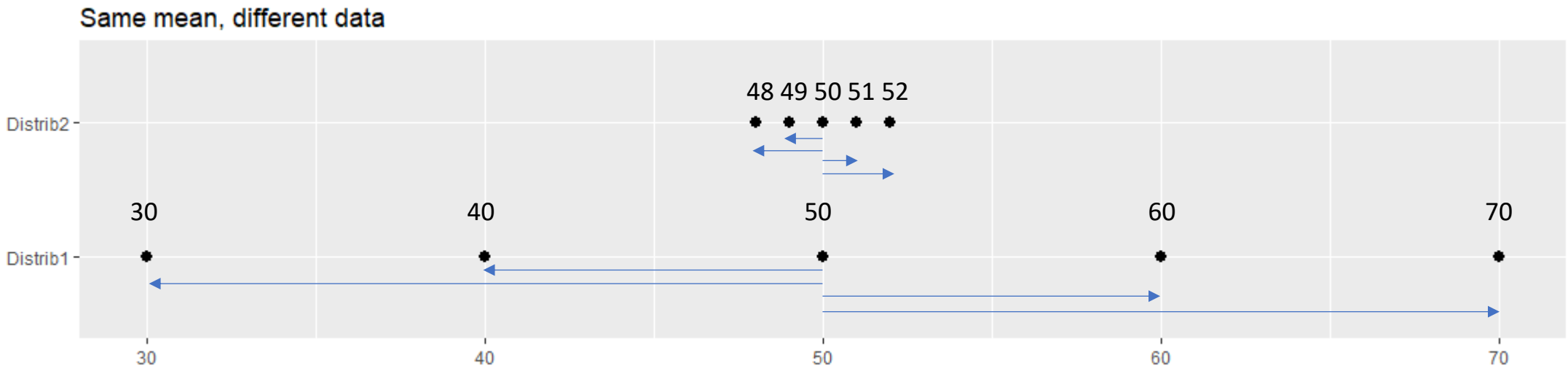
Measures of dispersion



$stDev_1 = 1.58$
$stDev_2 = 15.8$

The **standard deviation** measures the spread of the data.

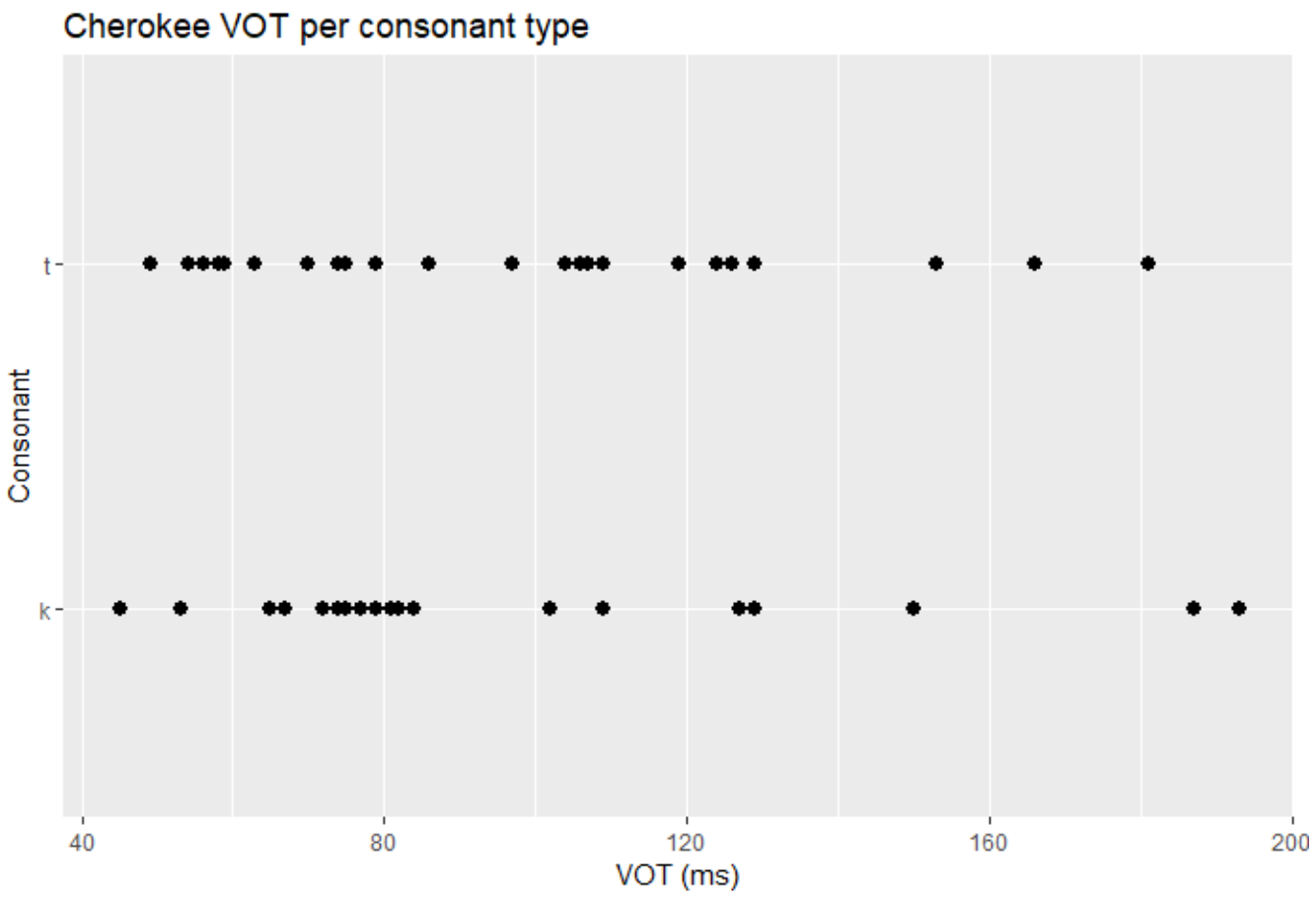
Measures of dispersion



$stDev_1 = 1.58$ $var_1 = 2.5$
$stDev_2 = 15.8$ $var_2 = 250$

The **variance** is the same, but squared.
The standard deviation is the most useful for linguistics, because it's in the same units as the original measurement.

Understanding the distribution of our data: Histograms

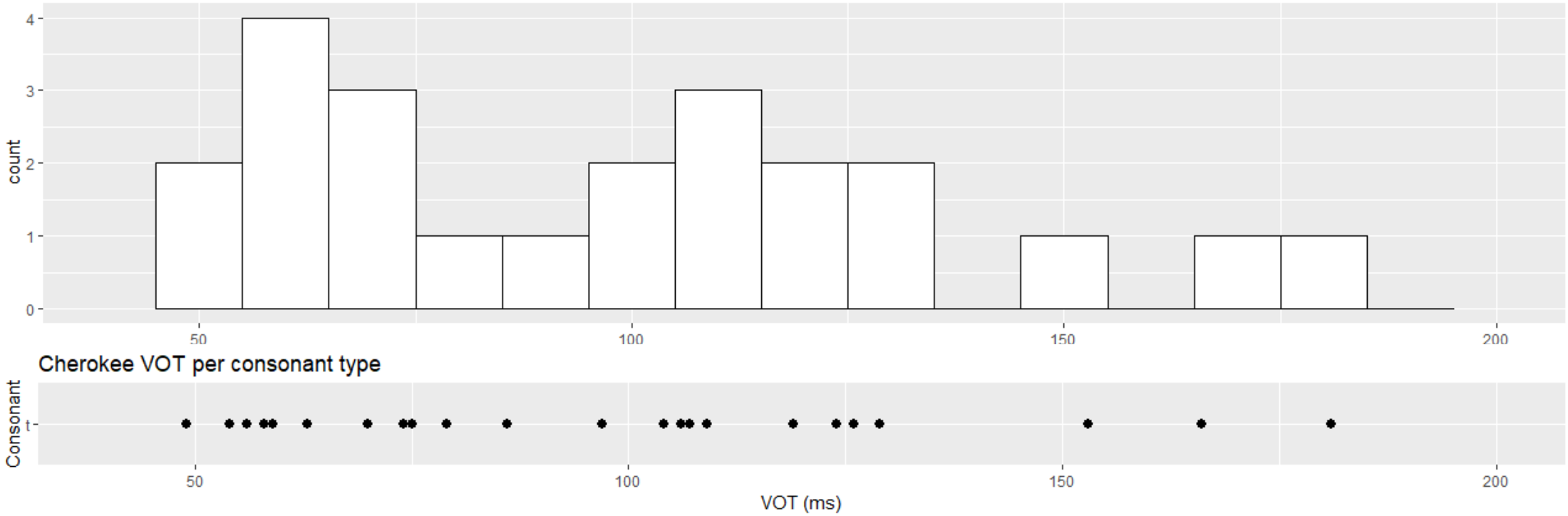


t: 49 54 56 58 59 63 70 74 75 79 86 97 104 106 107 109 119 124 126 129 153 166 181

k: 45 53 65 67 72 72 74 75 77 77 79 81 82 84 102 109 127 129 150 187 193

Understanding the distribution of our data: Histograms

t: [49 54] [56 58 59 63] [70 74 75] [79] [86] [97 104] [106 107 109] [119 124] [126 129] [153] [166] [181]



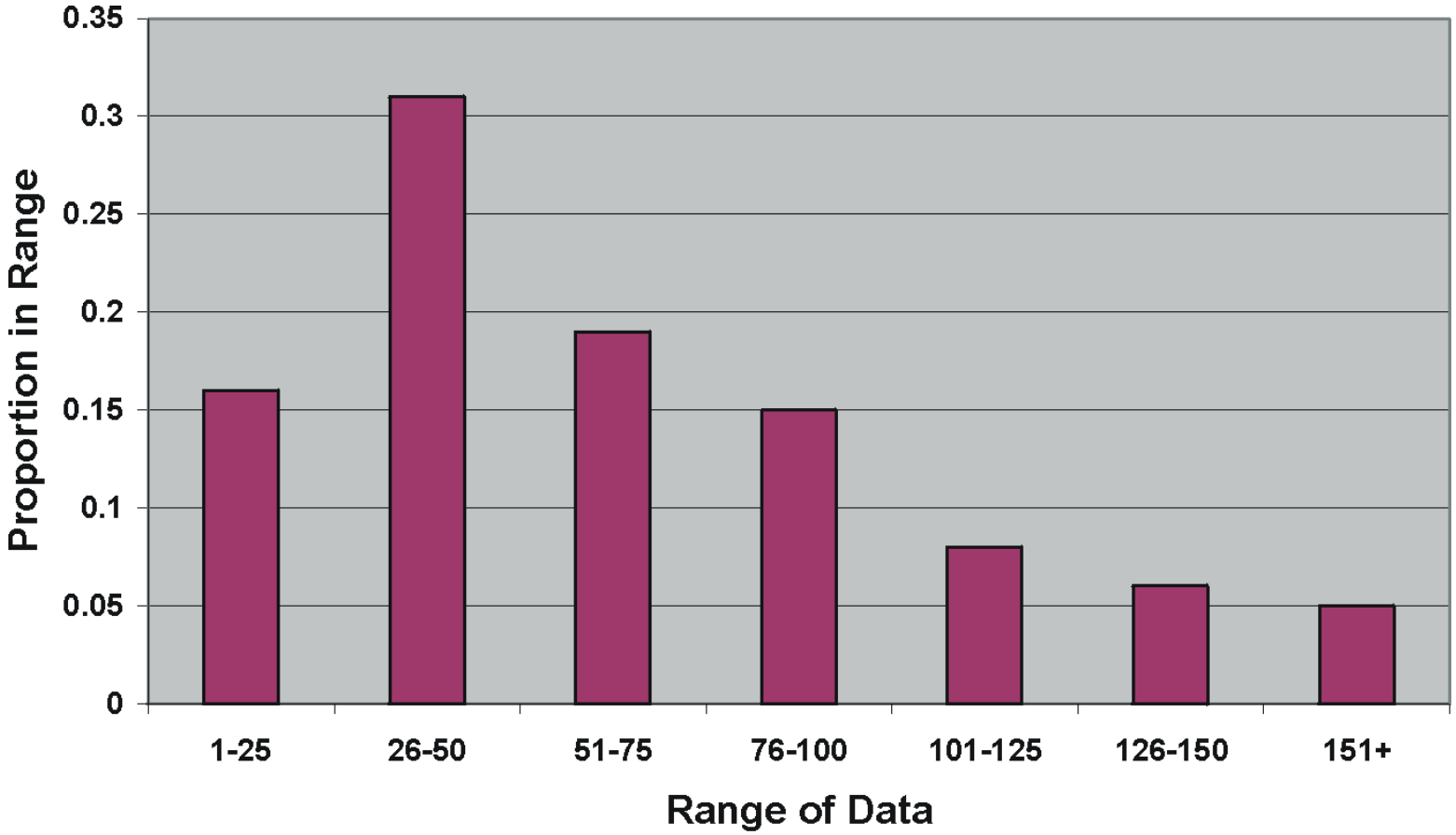
Properties of distributions

You should always try to look at your data before you begin your stats. Look for:

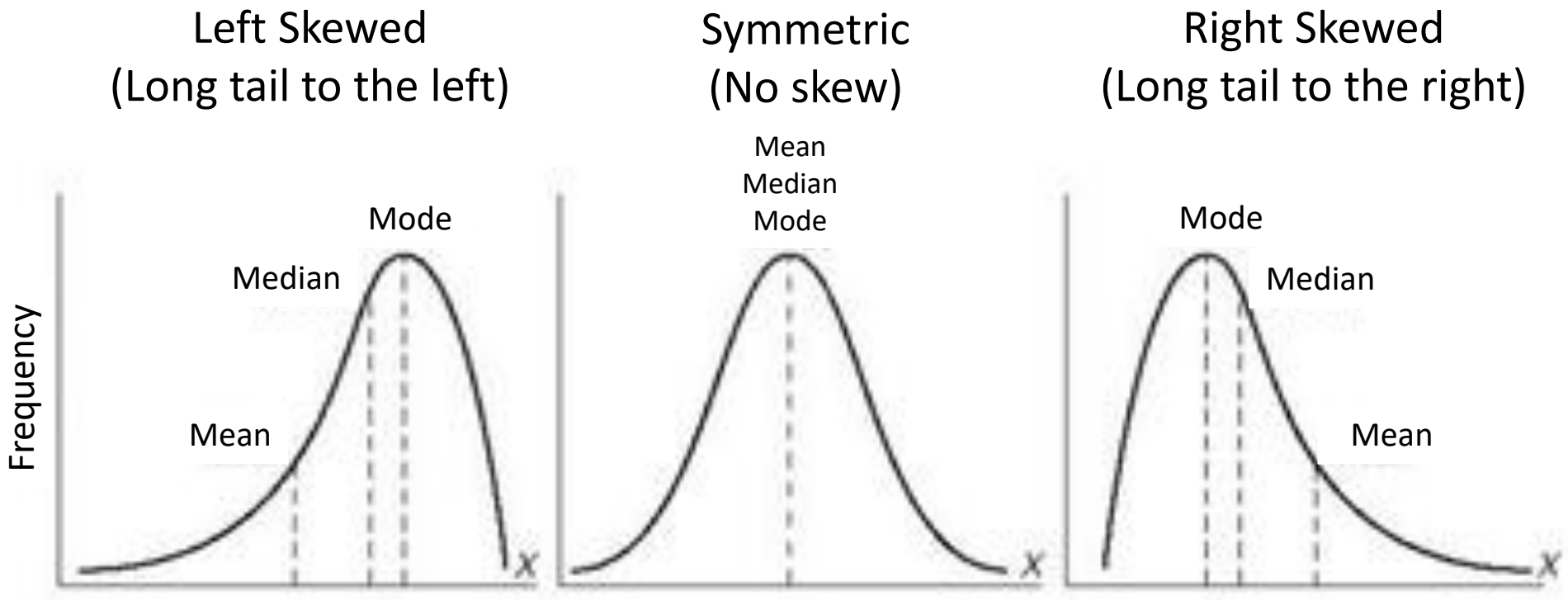
Range
Symmetry
Skew
Number and location of modes
Outliers

Range

The **range** is the minimum and maximum of values in the histogram (e.g. 1-200).

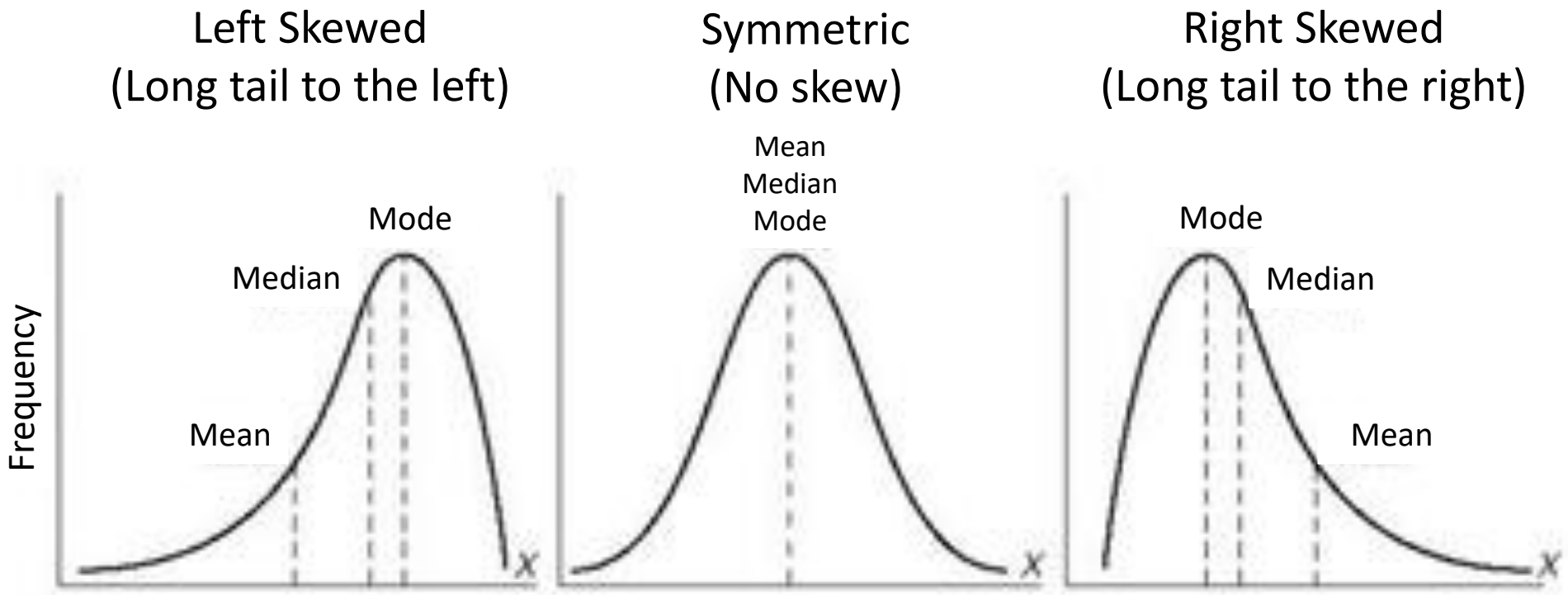


Symmetry and Skew



Histograms can be **symmetric** or assymetric.
If they are assymetric, then they are **skewed**.

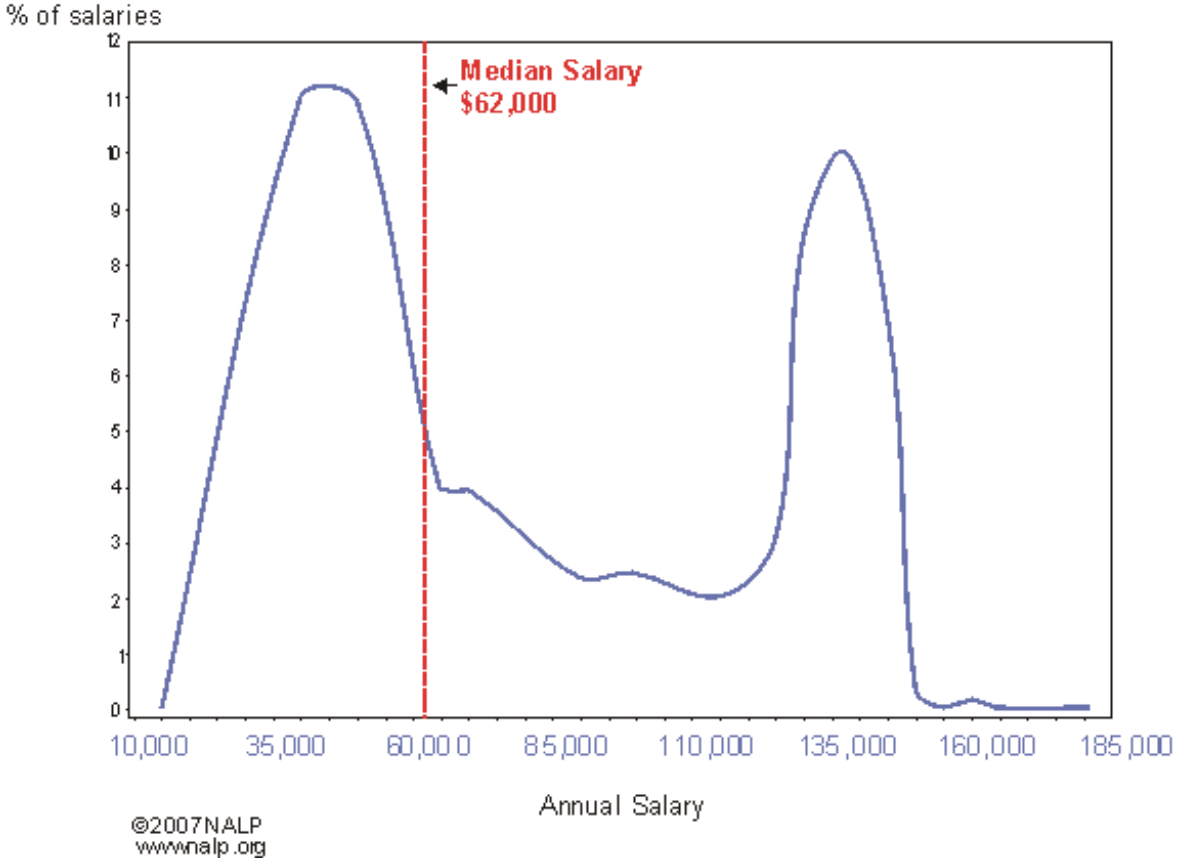
Symmetry and Skew



In histograms, the **mode** is the most frequent value.

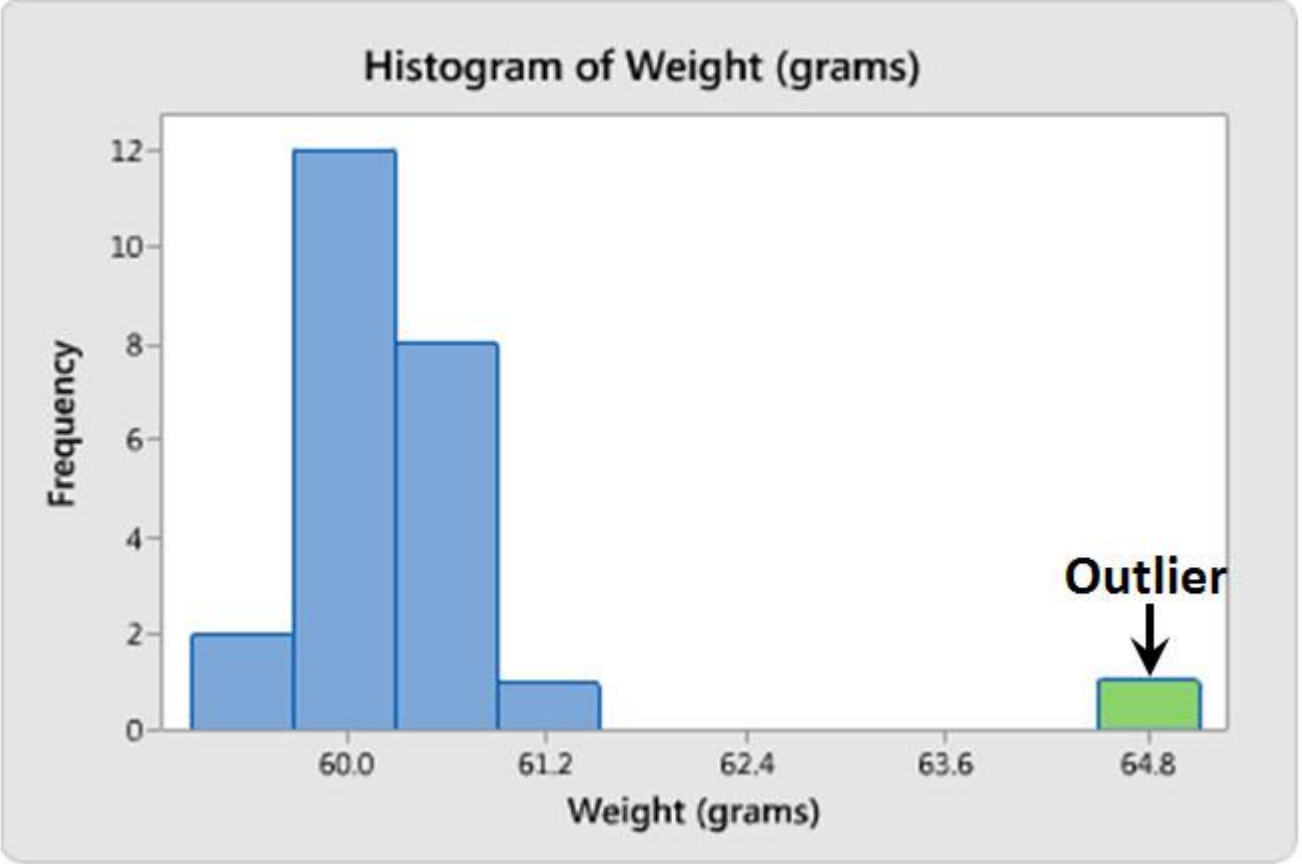
Unimodal and Bimodal Distributions

Some histograms can have more than one mode. When they have two, this distribution is known as **bimodal**. (When you have only one mode, this is a **unimodal** distribution).



Outliers

Outliers are values that seem out of the ordinary.



Outliers

Anna J. Egalite
@annaegalite

In my intro stats class today, I told students the median is a "resistant" measure of a distribution's center & is often preferred to the mean in the case of salary data, etc. I jokingly referenced this meme and in the 15 mins' break they had, a student created this MASTERPIECE!

Standard Deviation, Variance and Distributions

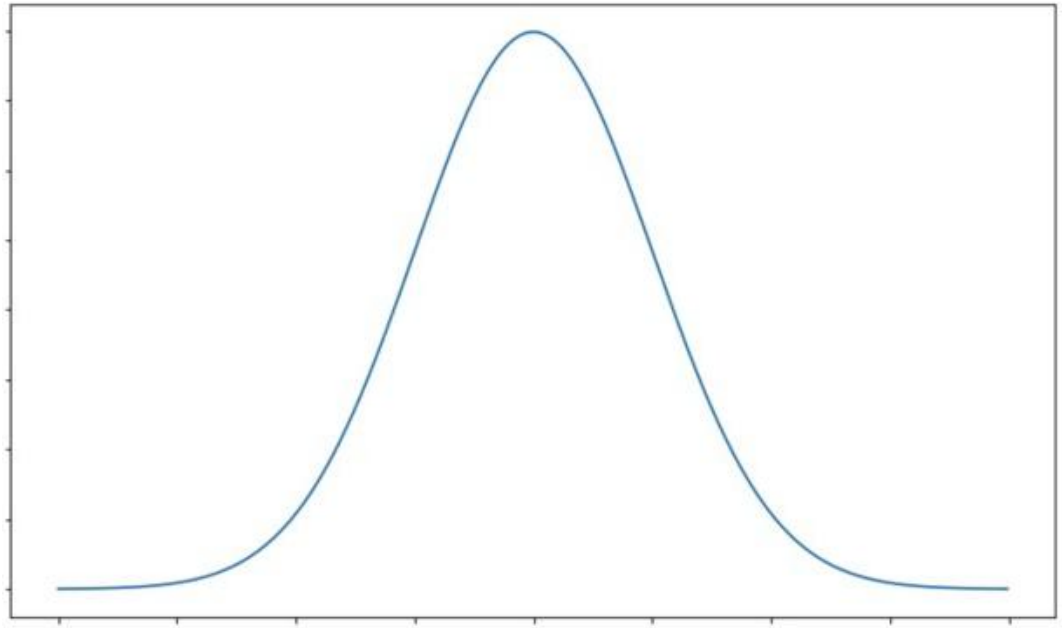
Imagine you have a histogram that looks like this.

How would it change…

(1) With a higher value for the mean?

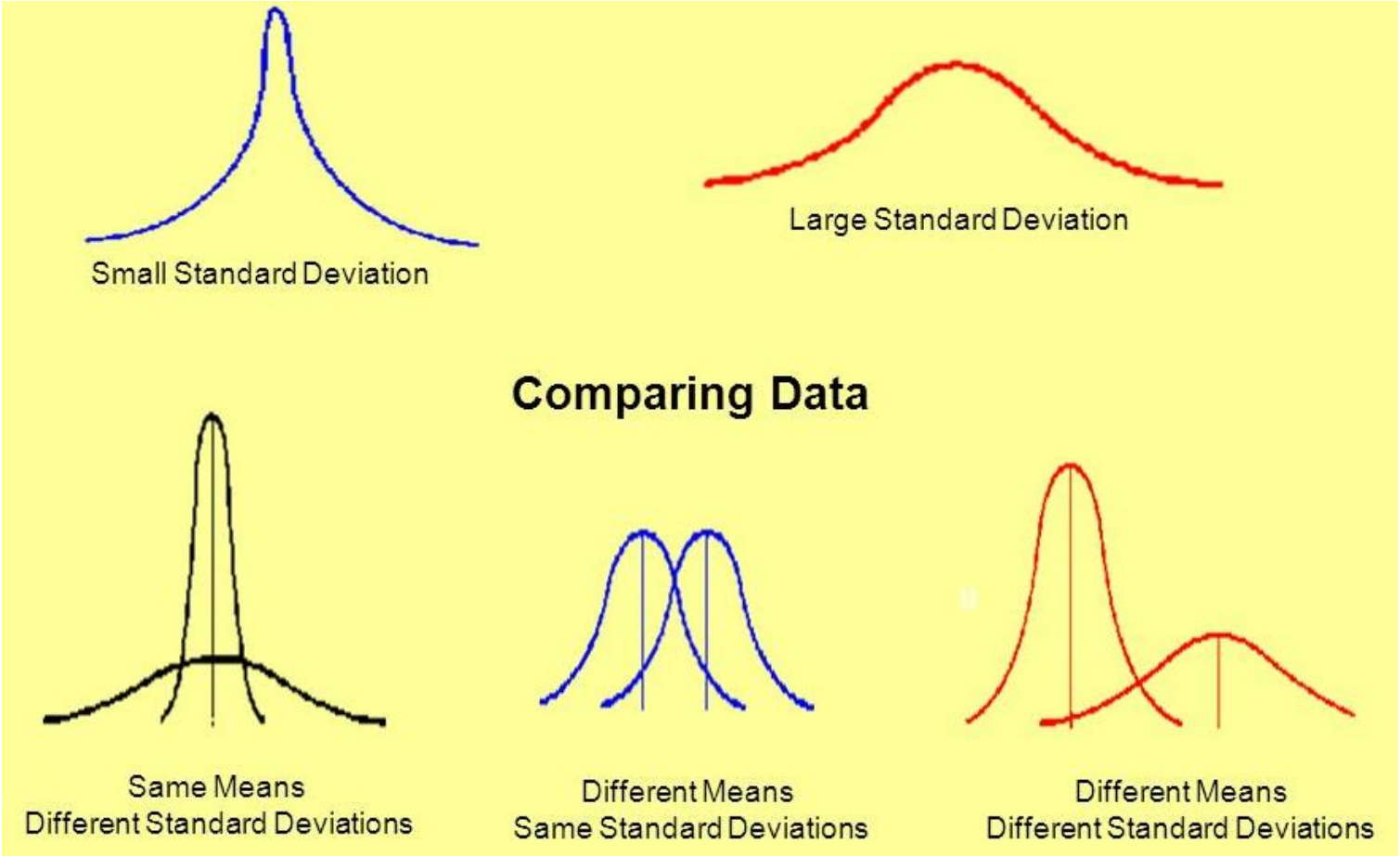
(2) With a higher standard deviation?

(3) With both a higher value for the mean and a higher standard deviation?



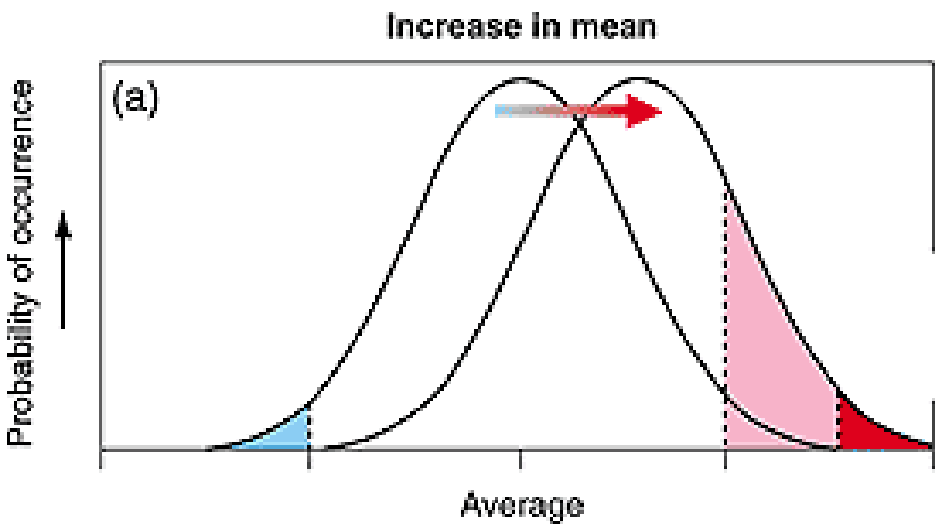
Standard Deviation, Variance and Distributions

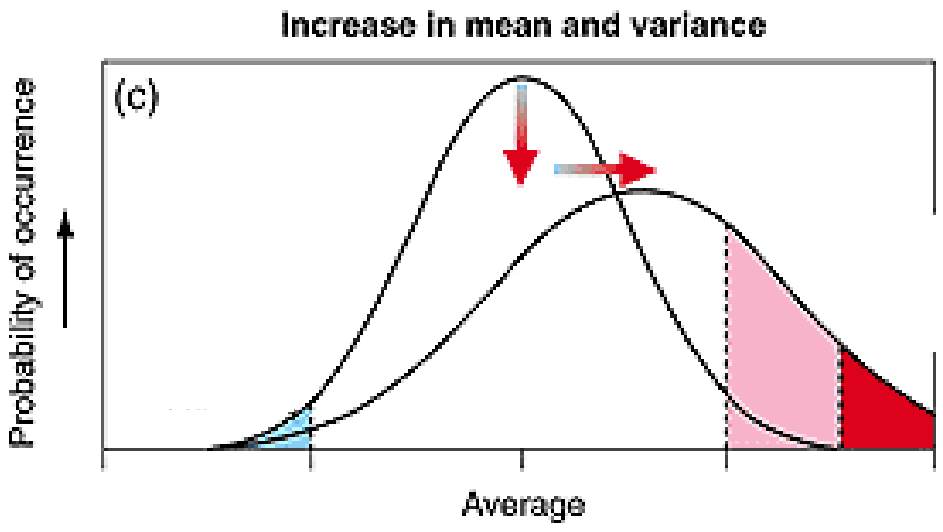
The **sample standard deviation** measures the spread of our data
It is a measure of average squared deviation (distance) from the mean

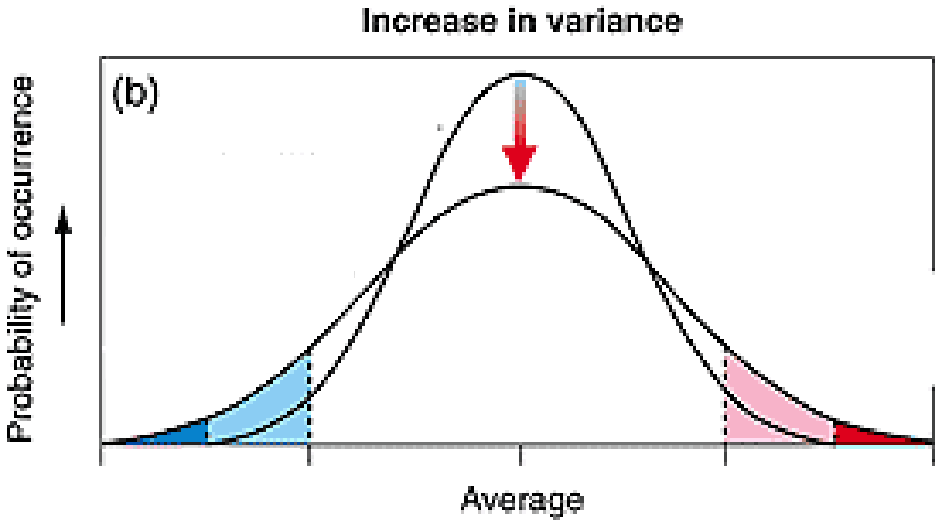


Peter Matthews

Standard Deviation, Variance and Datasets

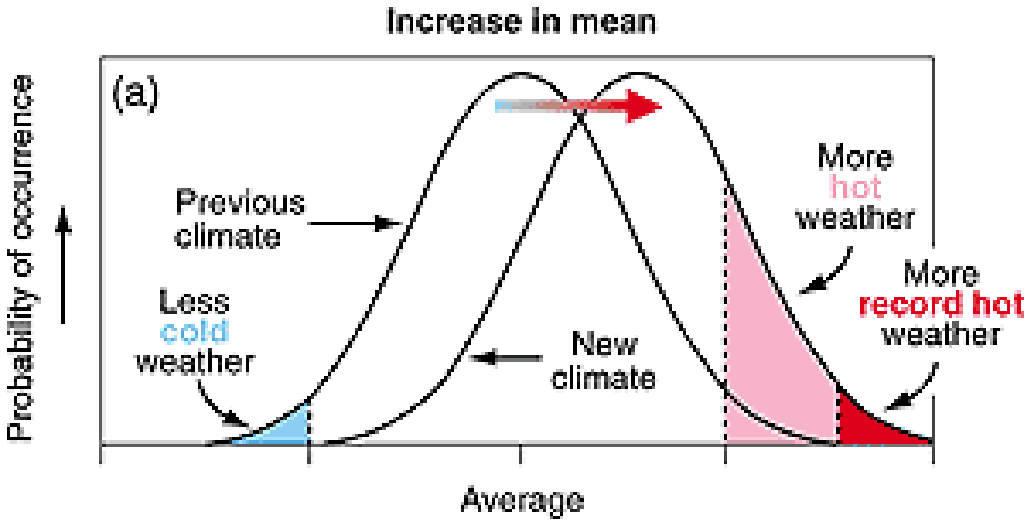


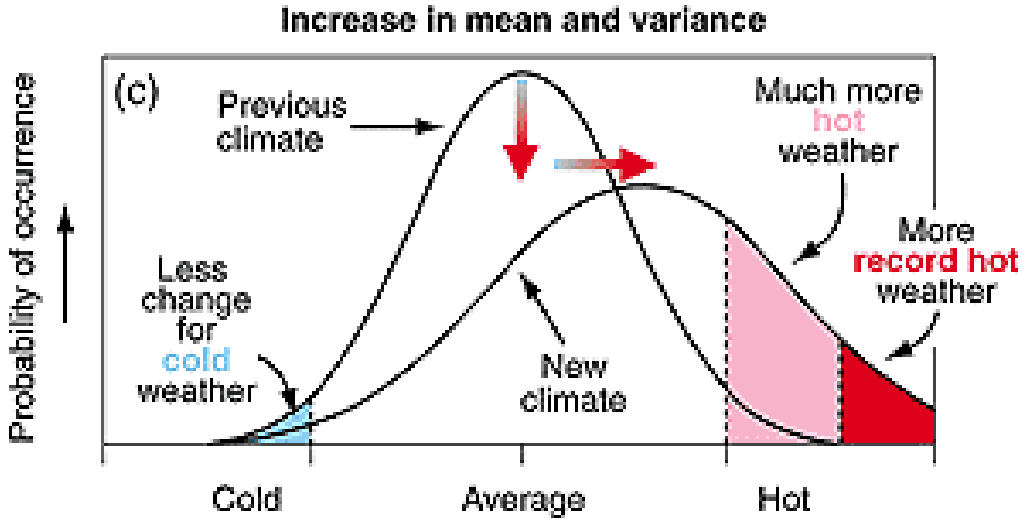


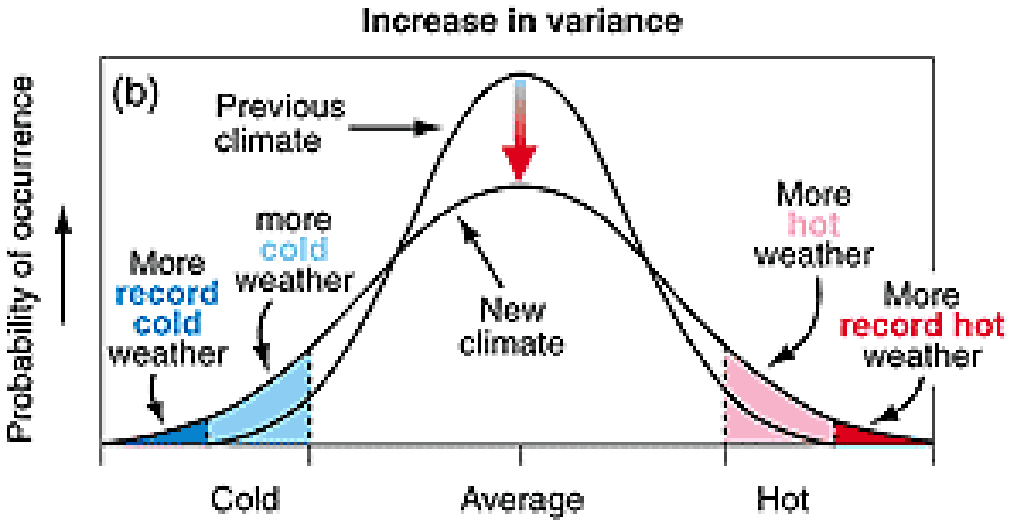


Source: IPCC (2001:155) >>

Standard Deviation, Variance and Datasets





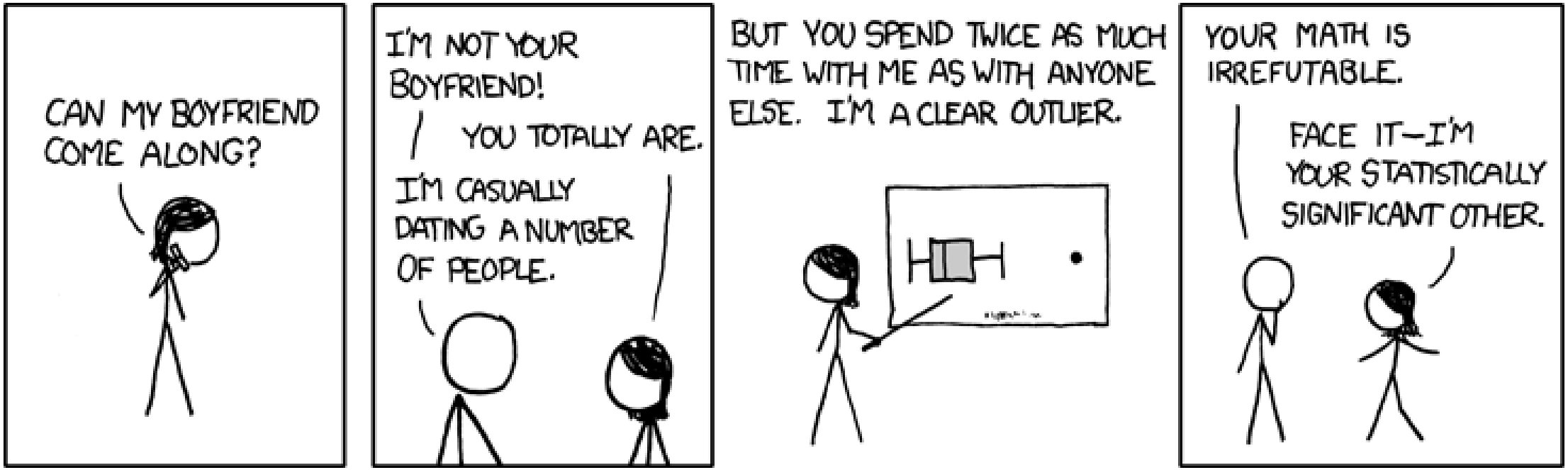


Source: IPCC (2001:155) >>

Summary

We need to understand the **dispersion** in our data. We can measure it with the standard deviation and the variance.

We can use a **histogram** to improve our understanding of the distribution of the data and detect phenomena like skew and outliers.

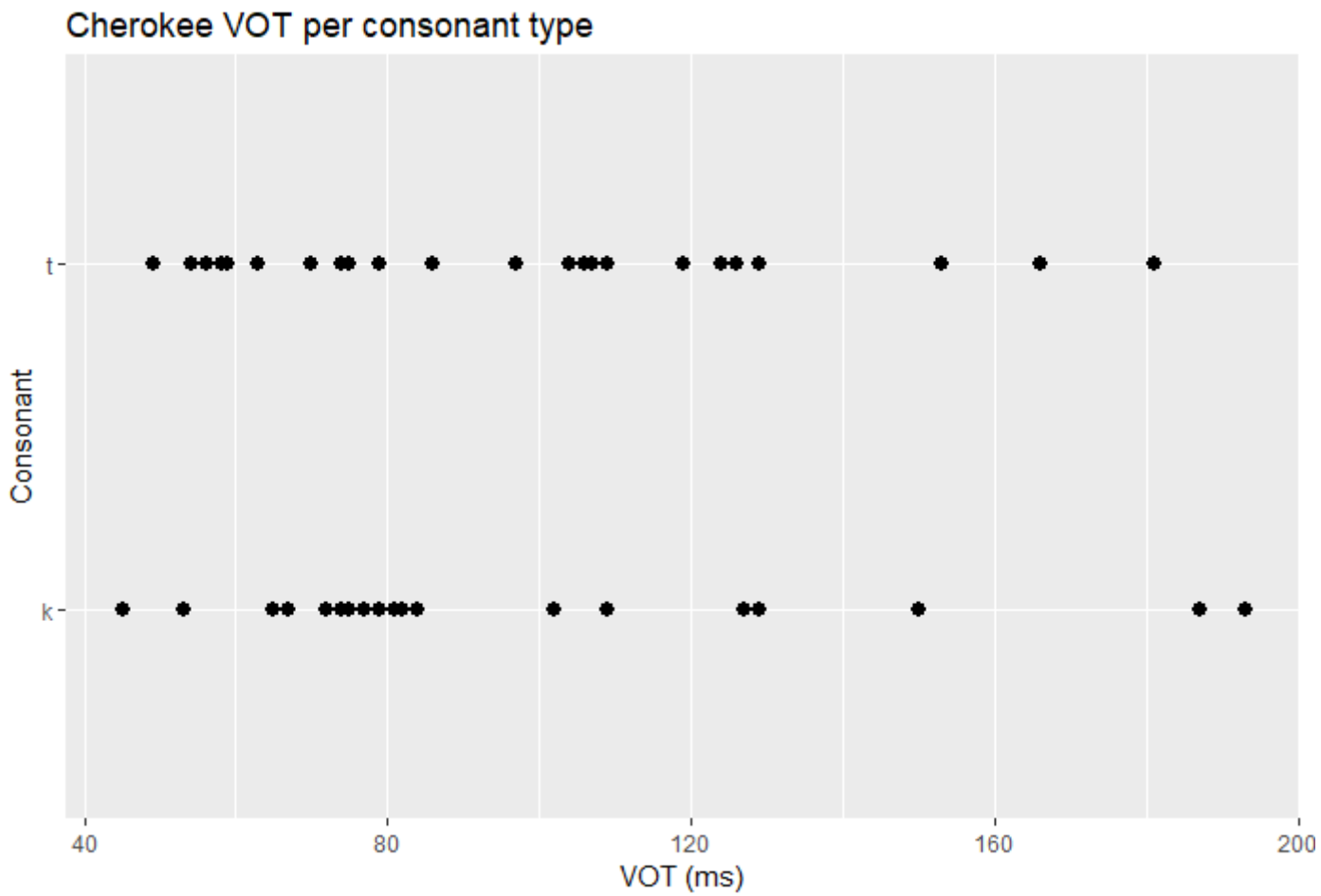


Quartiles and Boxplots

Rolando Coto-Solano. Dartmouth College

LING10, January 2026

Understanding the distribution of our data: Histograms



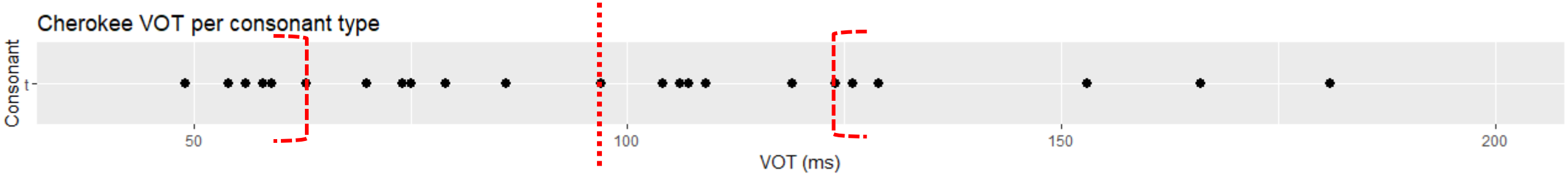
t: 49 54 56 58 59 63 70 74 75 79 86 97 104 106 107 109 119 124 126 129 153 166 181

k: 45 53 65 67 72 72 74 75 77 77 79 81 82 84 102 109 127 129 150 187 193

Quartiles

25% of the data
We have 50% so far

25% of the data
We have 100%



25% of the data
We have 25% so far

Median

25% of the data
We have 75% so far

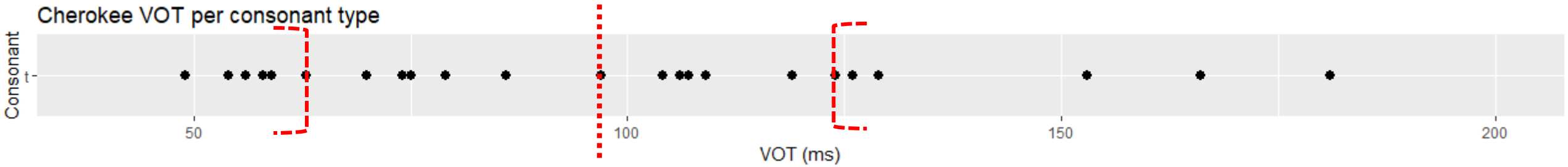
t: 49 54 56 58 59 63 70 74 75 79 86 **97** 104 106 107 109 119 124 126 129 153 166 181

Median

Quartiles

2nd quartile
97.0

4th quartile:
181.0



1st quartile
"Lower" quartile
66.5

Median

3rd quartile
"Upper" quartile
121.5

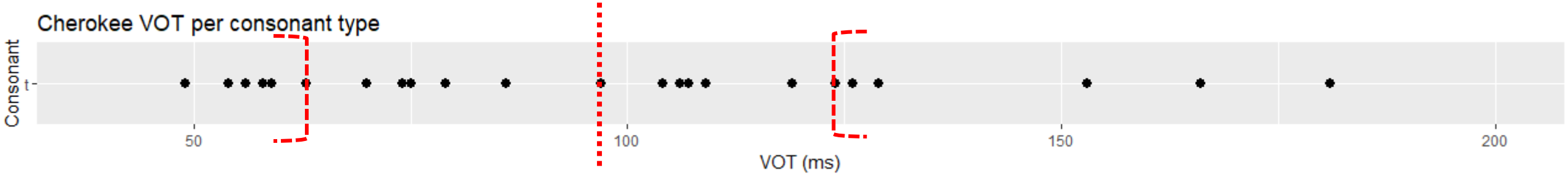
t: 49 54 56 58 59 63 70 74 75 79 86 **97** 104 106 107 109 119 124 126 129 153 166 181

Median

Quartiles

2nd quartile
97.0

4th quartile:
181.0



1st quartile
"Lower" quartile
66.5

Median

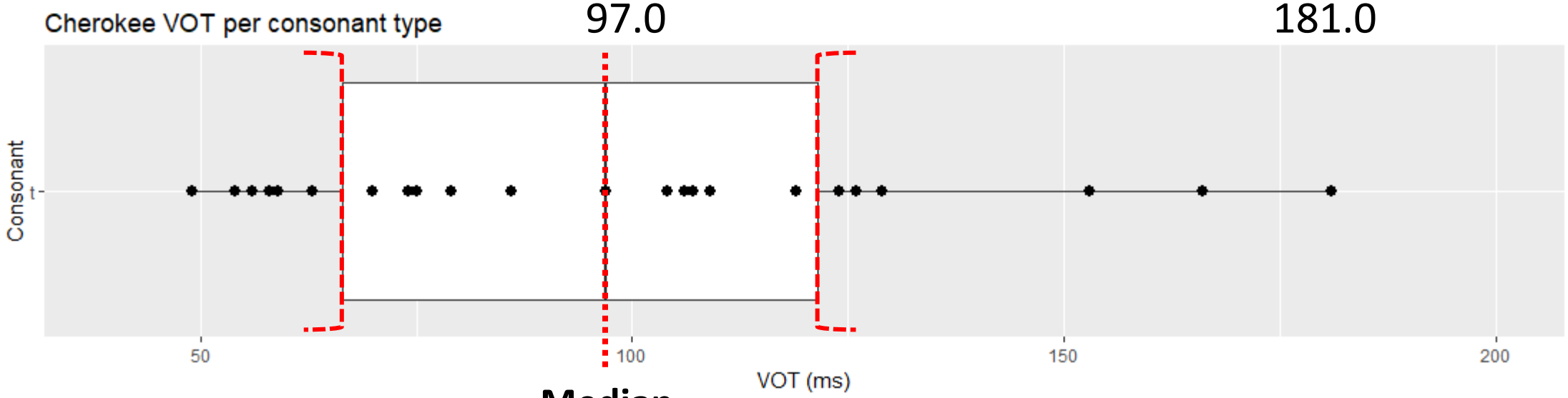
3rd quartile
"Upper" quartile
121.5

Interquartile range:
121.5 – 66.5 = 55

Quartiles and Boxplots

2nd quartile
97.0

4th quartile:
181.0



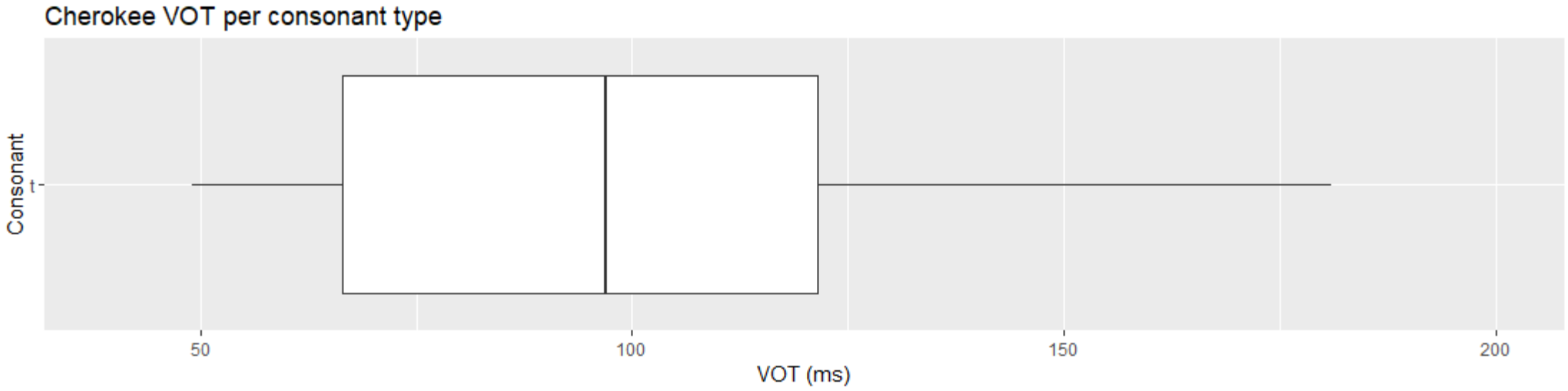
Median

1st quartile
"Lower" quartile
66.5

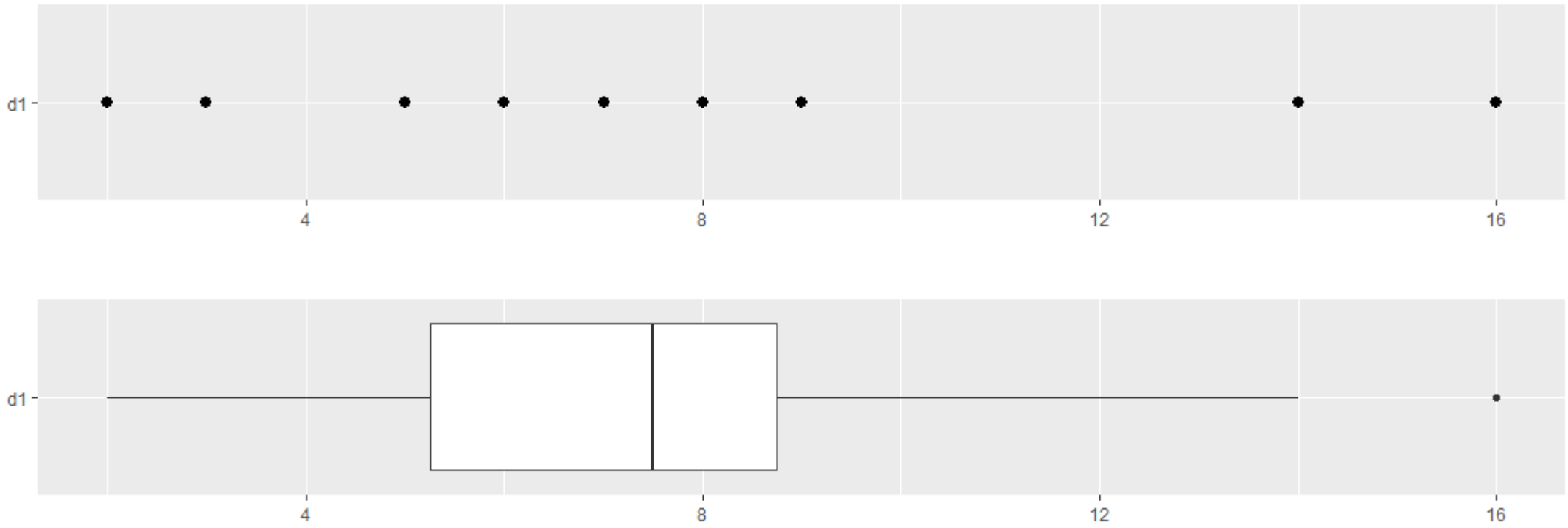
3rd quartile
"Upper" quartile
121.5

Interquartile range:
121.5 – 66.5 = 55

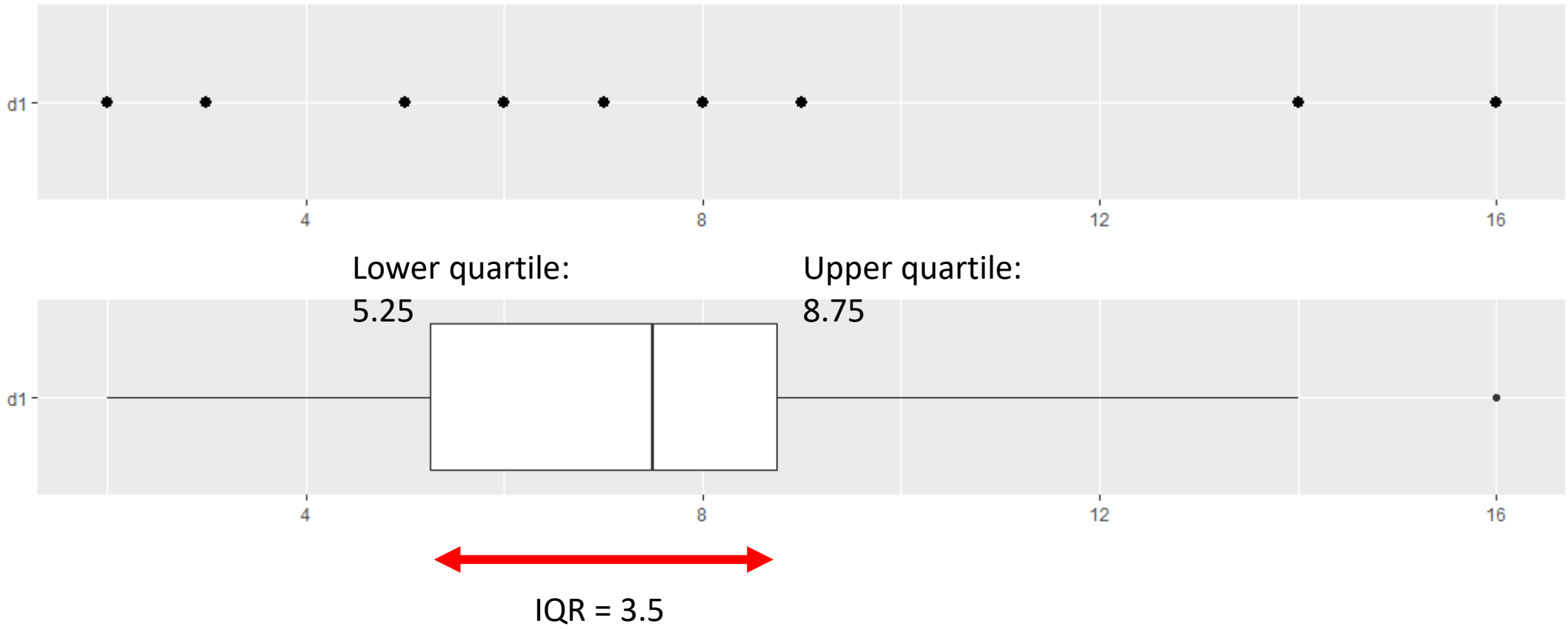
Boxplots


Cherokee VOT per consonant type
Consonant
t
50
100
150
200
VOT (ms)

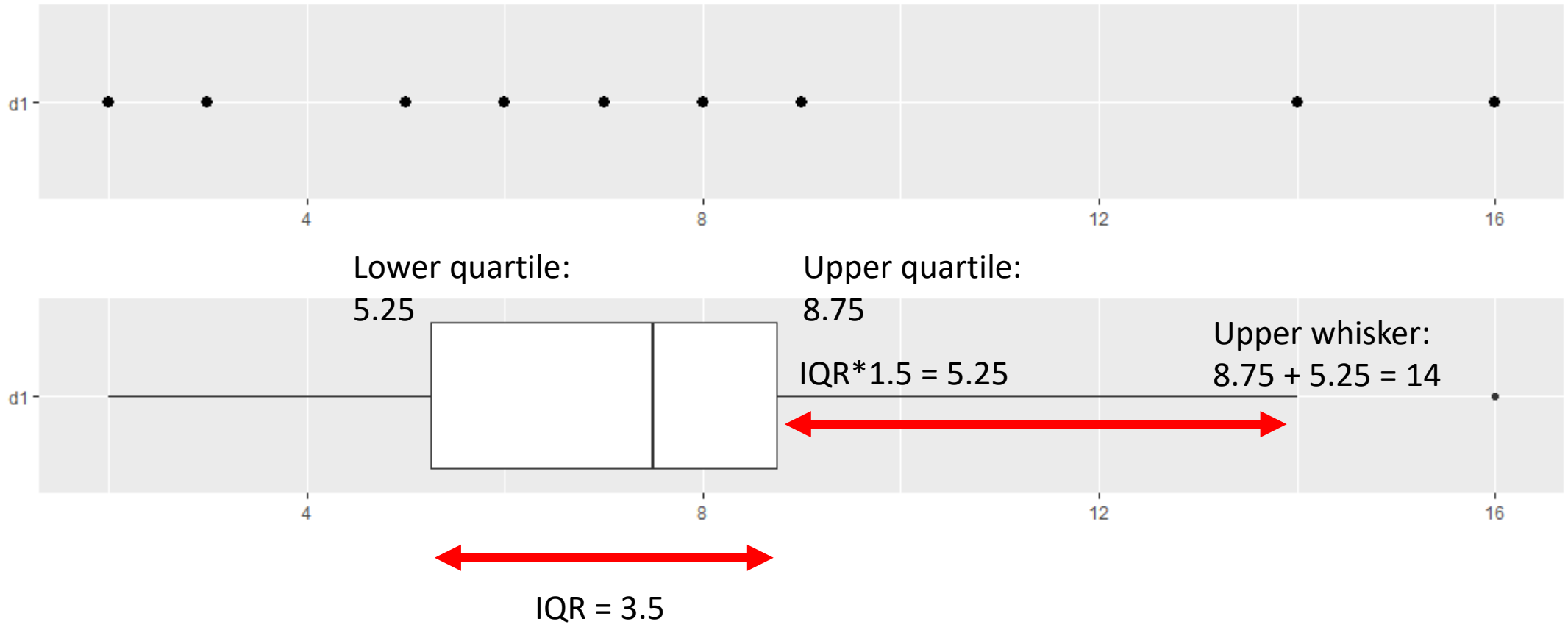
Boxplot Whiskers and Outliers



Boxplot Whiskers and Outliers

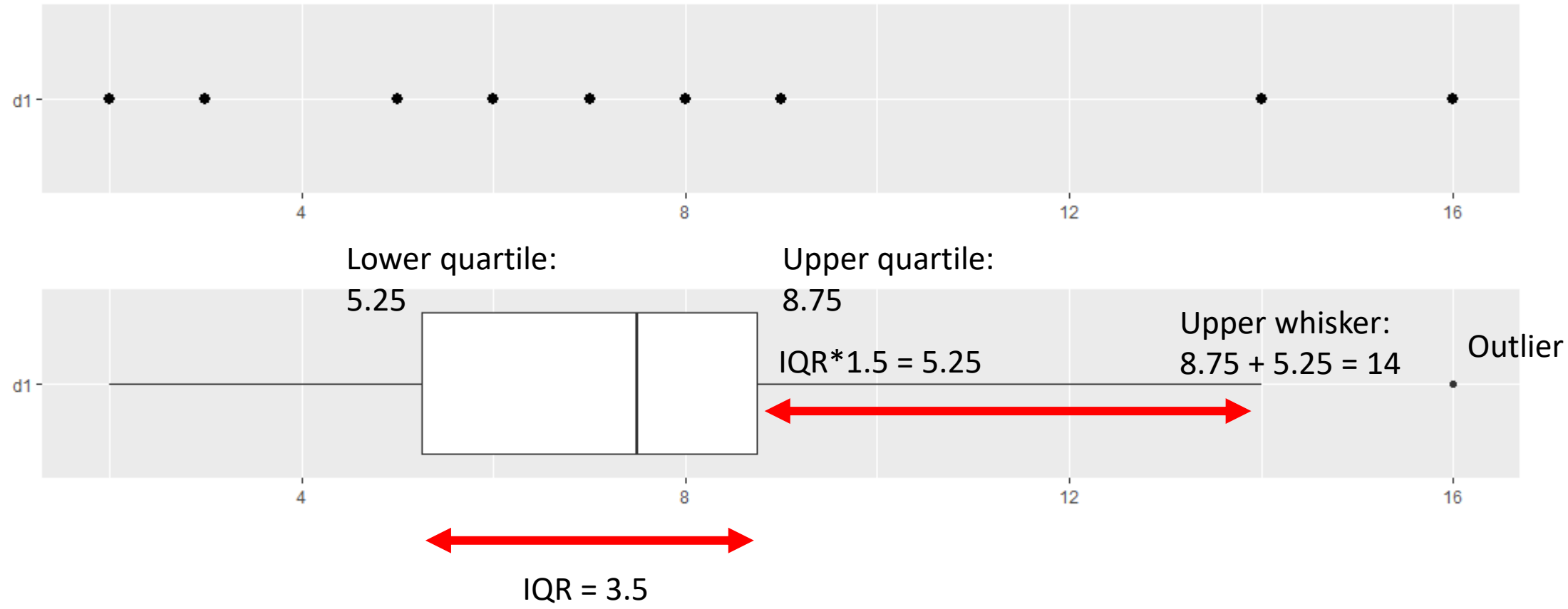


Boxplot Whiskers and Outliers



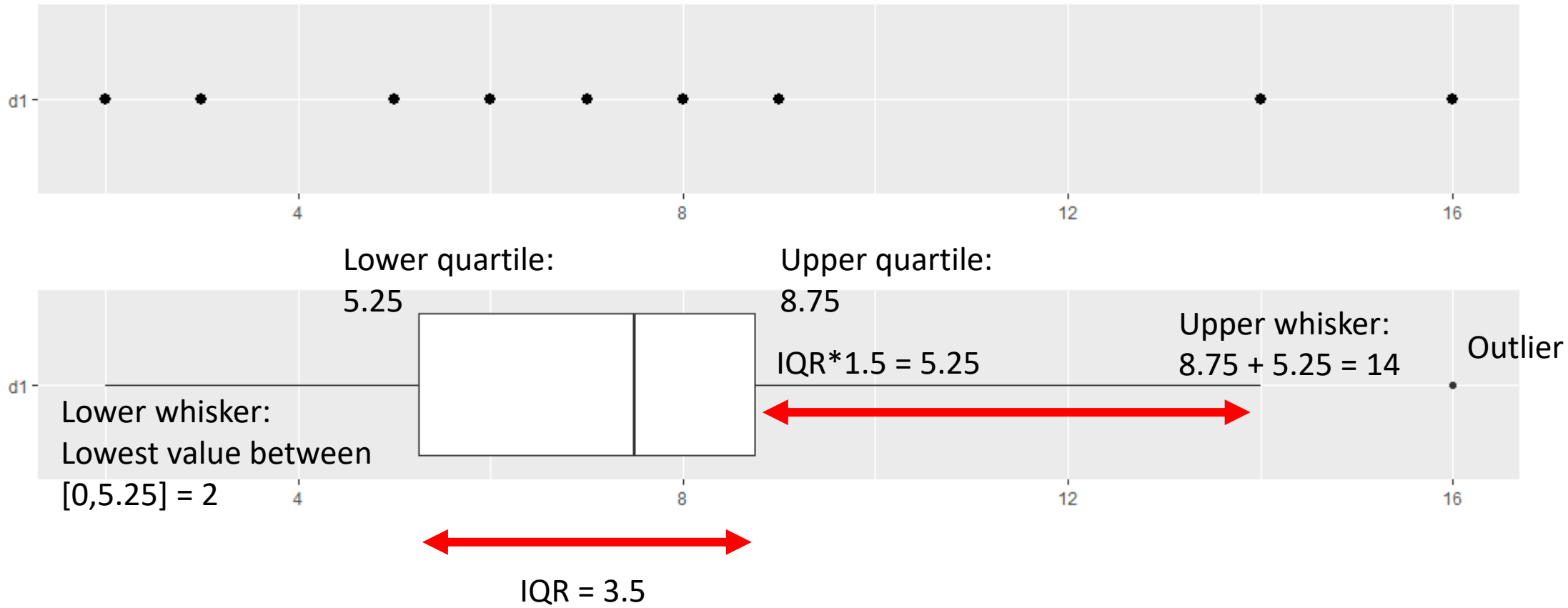
"The upper whisker extends from the hinge to the highest value that is within 1.5 * IQR of the hinge, where IQR is the inter-quartile range, or distance between the first and third quartiles. The lower whisker extends from the hinge to the lowest value within 1.5 * IQR of the hinge. Data beyond the end of the whiskers are outliers and plotted as points" (R documentation)

Boxplot Whiskers and Outliers



"The upper whisker extends from the hinge to the highest value that is within 1.5 * IQR of the hinge, where IQR is the inter-quartile range, or distance between the first and third quartiles. The lower whisker extends from the hinge to the lowest value within 1.5 * IQR of the hinge. Data beyond the end of the whiskers are outliers and plotted as points" (R documentation)

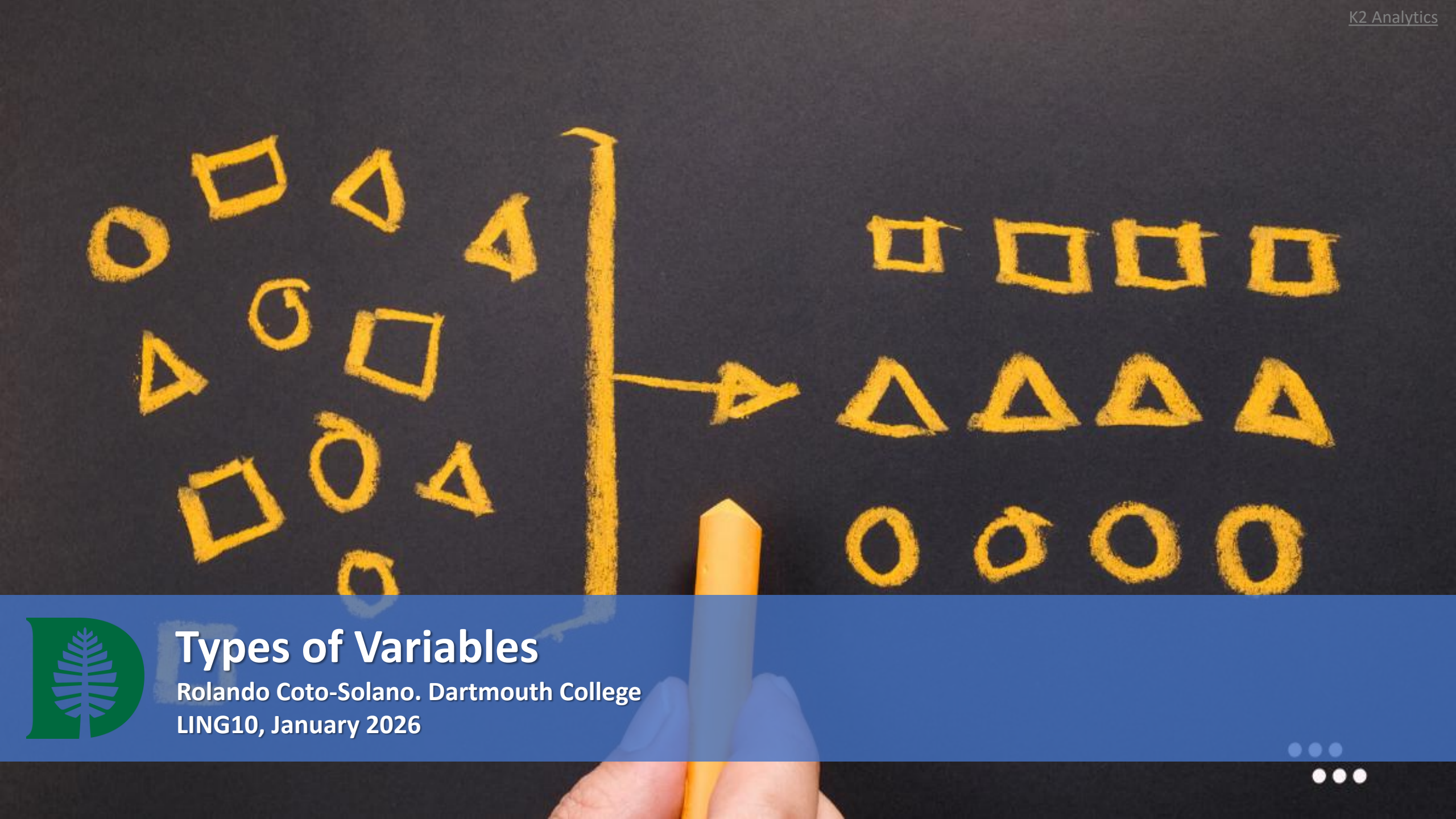
Boxplot Whiskers and Outliers



"The upper whisker extends from the hinge to the highest value that is within 1.5 * IQR of the hinge, where IQR is the inter-quartile range, or distance between the first and third quartiles. The lower whisker extends from the hinge to the lowest value within 1.5 * IQR of the hinge. Data beyond the end of the whiskers are outliers and plotted as points" (R documentation)

Summary

Boxplots are a quick way to understand the range and distribution of the data, as well as the presence of outliers.

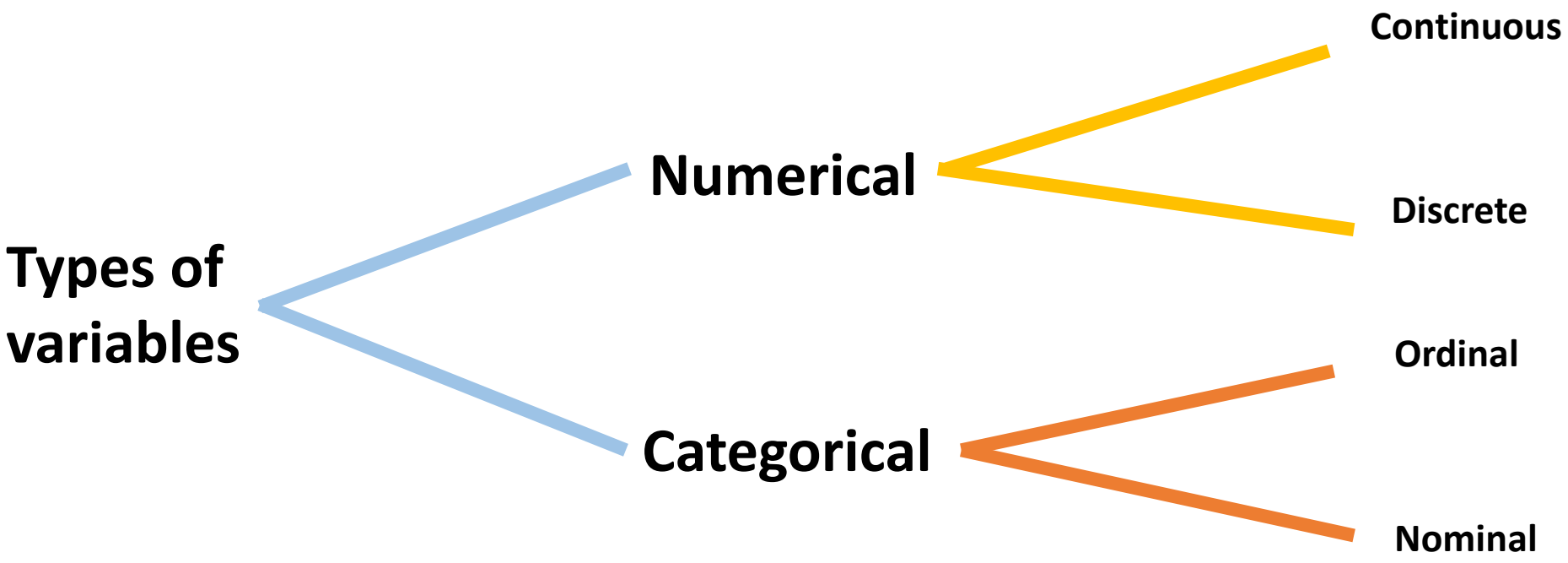


Types of Variables

Rolando Coto-Solano. Dartmouth College

LING10, January 2026

Types of Variables



Types of Variables

Types of variables

- **Numerical**
  - **Continuous** (e.g. measurements)
  - **Discrete** (e.g. how many…?)
- **Categorical**
  - **Ordinal**
  - **Nominal**

Types of Variables

Types of variables

- **Numerical**
  - **Continuous** (e.g. measurements)
  - **Discrete** (e.g. how many…?)
- **Categorical**
  - **Ordinal** (when categories have a logical order e.g. stress level, approval rating for ..)
  - **Nominal** (e.g. nameSchool, favorite color)

Identifying Variable Types

Consider the description of the variable:
Does the variable represent categories (categorical) or measurements (numerical)?

If **numerical** – consider the description of the variable

What is the range of values the variable can take? (identify minimum and maximum)

Are all values between the minimum and maximum possible for the variable **as it is described?** (it may help to draw a number line)

Yes	**Continuous (all values possible)**
No	**Discrete (only a fixed set possible)**

If **categorical** – consider the categories

Do the categories follow a natural order?

Yes	**Ordinal**
No	**Nominal**

Identifying Variable Types

Discrete data may be, **but are not always**, whole numbers.

They are numbers which can be given exactly, without the need for rounding. For example

how many pets do you have? {0, 1, 2, 3…}

shoe sizes (interval between numbers could be 0.5) as in {…36, 36.5, 37, 37.5…}

Identifying Variable Types

Continuous data can in theory take any numerical value within a certain range. They may be rounded for convenience. For example

how old are you?
20 years, or 20 years 4 months, or 20 years 4 months 3 days, 18 hours…

average yearly income
50000, 50308, 50308.24…

how many hertz in the formant of the vowel
105, 105.1, 105.14, 105.139, 105.1387…

What is the type of variable for each of these?

	speaker.id	age	word	arpabet.word	vowel	stressed	F1
1	h01	31	hello	HH AH0 L OW1	AH	0	520
2	h01	31	hello	HH AH0 L OW1	OW	1	371
3	h01	31	how	HH AW1	AW	1	763
4	h01	31	are	AA1 R	AA	1	849
5	h01	31	you	Y UW1	UW	1	320
6	h02	26	hello	HH AH0 L OW1	AH	0	531
7	h02	26	hello	HH AH0 L OW1	OW	1	368
8	h02	26	how	HH AW1	AW	1	781
9	h02	26	are	AA1 R	AA	1	833
10	h02	26	you	Y UW1	UW	1	309

Numerical: Continuous
Discrete

Categorical: Ordinal
Nominal

What is the type of variable for each of these?

	speaker.id	age	word	arpabet.word	vowel	stressed	F1
1	h01	31	hello	HH AH0 L OW1	AH	0	520
2	h01	31	hello	HH AH0 L OW1	OW	1	371
3	h01	31	how	HH AW1	AW	1	763
4	h01	31	are	AA1 R	AA	1	849
5	h01	31	you	Y UW1	UW	1	320
6	h02	26	hello	HH AH0 L OW1	AH	0	531
7	h02	26	hello	HH AH0 L OW1	OW	1	368
8	h02	26	how	HH AW1	AW	1	781
9	h02	26	are	AA1 R	AA	1	833
10	h02	26	you	Y UW1	UW	1	309

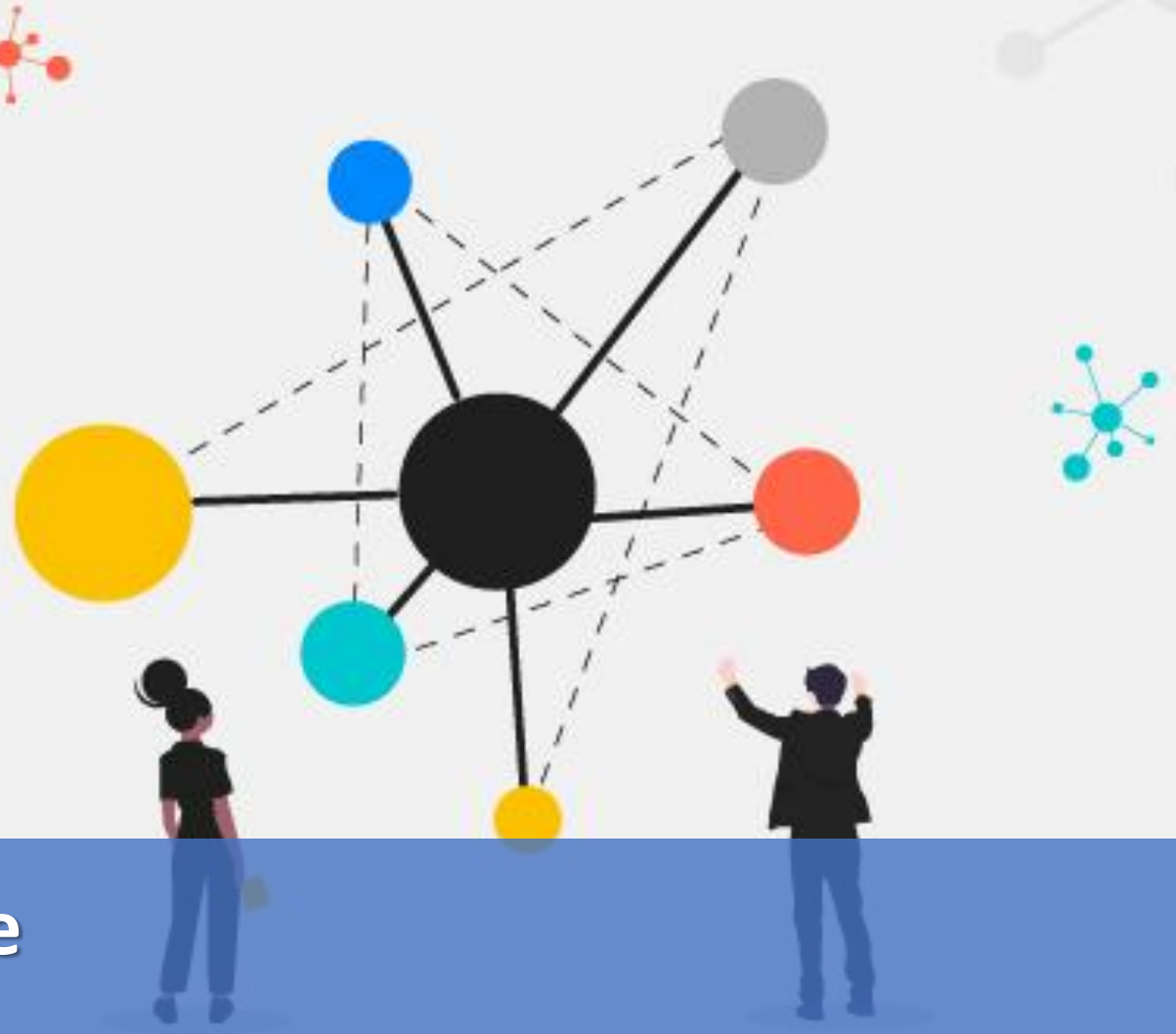
speaker.id: Categorical, nominal
age: Numerical, continuous(?)
word: Categorical, nominal
arpabet.word: Categorical, nominal
vowel: Categorical, nominal
stressed: Numerical, discrete
F1: Numerical, continuous

Numerical: Continuous
Discrete
Categorical: Ordinal
Nominal

Summary

There are different types (numerical, categorical) that we can use to describe our data.

Innoplexus



How to describe a variable

Rolando Coto-Solano. Dartmouth College

LING10, January 2026

Critical Thinking: What's in a variable

Some important branches of philosophy are:

Epistemology

How do we know things are real?
What evidence do we admit before conceding that something is real?

Axiology

Why do we think some things are beautiful, valuable, or worthy of our time?

Ontology

How do we structure the world?
What are the categories that we use to explain the world?

Critical Thinking: What's in a variable

Ontology

How do we structure the world?

What are the categories that we use to explain the world?

B What is your sex?

- Please select the option for how you identify.
- Tick **one** box only

☐ Female

☐ Male

☐ Other, please write in

Scotland Census 2021

https://www.scotlandscensus.gov.uk/documents/census2021/Sex_and_Gender_Identity_Topic_Report.pdf

Critical Thinking: What's in a variable

Ontology

How do we structure the world?

What are the categories that we use to explain the world?

© copyright Commonwealth of Australia (Geoscience Australia) [2006]

"Indigenous"

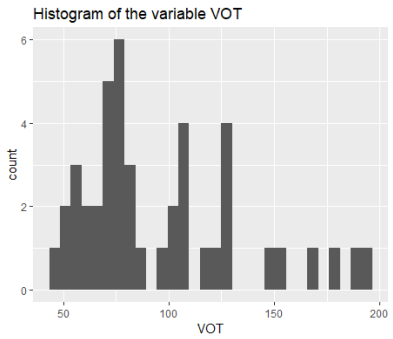
(see Walter & Andersen (2013). *Indigenous Statistics*).

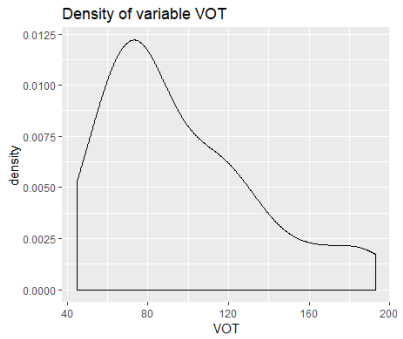
Critical Thinking: What’s in a variable

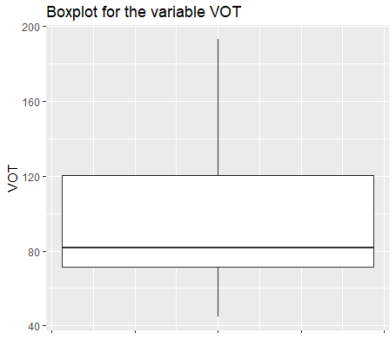
Whenever we categorize the world (i.e. create a categorical variable), we run the risk of forcefully fitting some data points into the categories we have chosen.

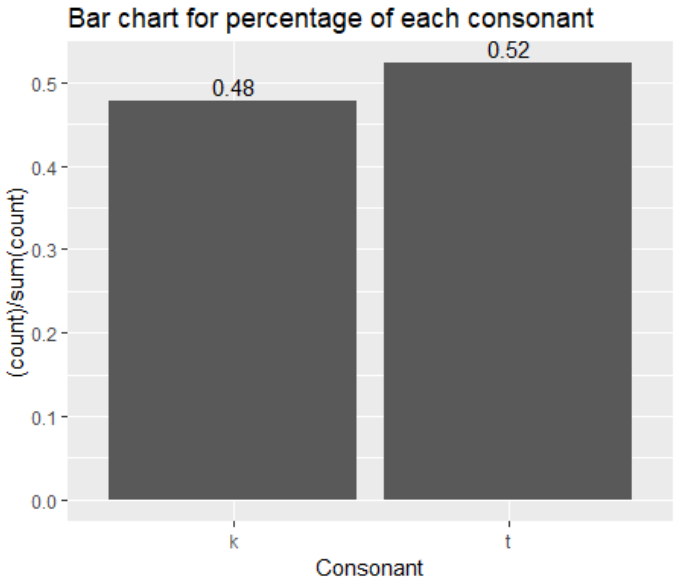
How to describe one variable?

Numerical	Categorical
Histogram	Barchart
Density plot	
Boxplot	





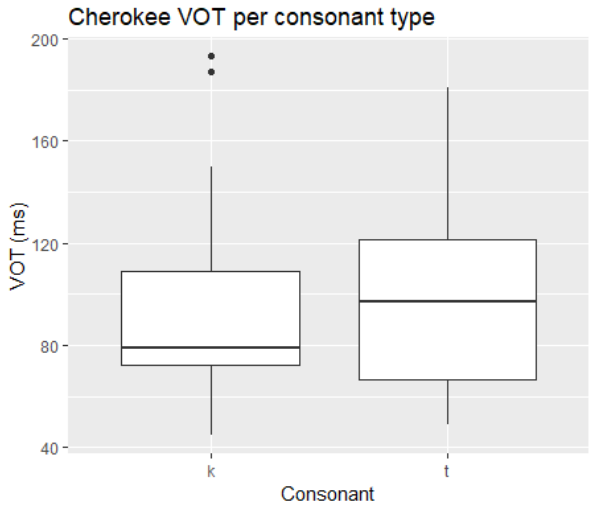




How to describe the relationship between two variables?

Categorical x continuous

Boxplot



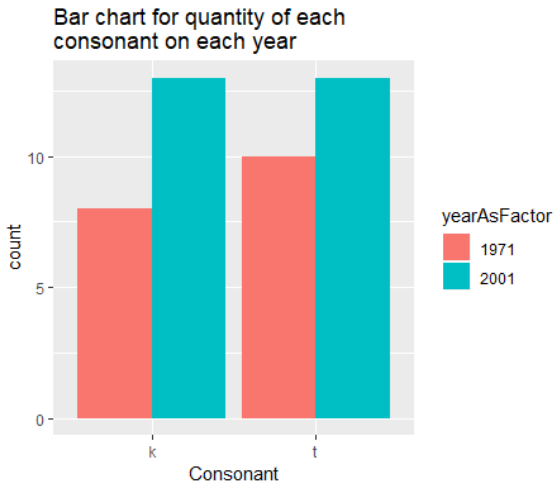
Categorical x categorical

Contingency table

Clustered barchart

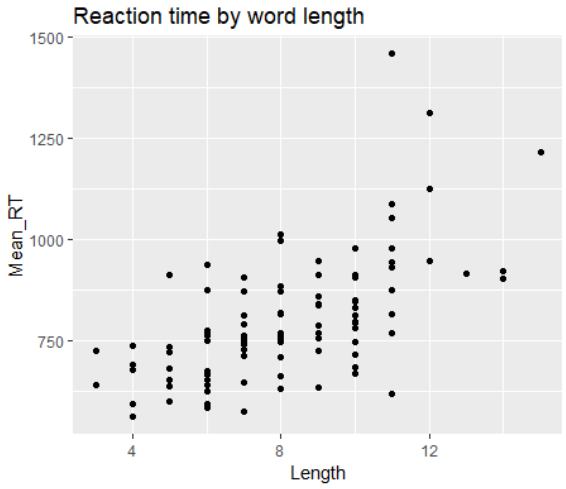
Cherokee consonants per year (in the sample)

	k	t
1971	8	10
2001	13	13



Continuous x continuous

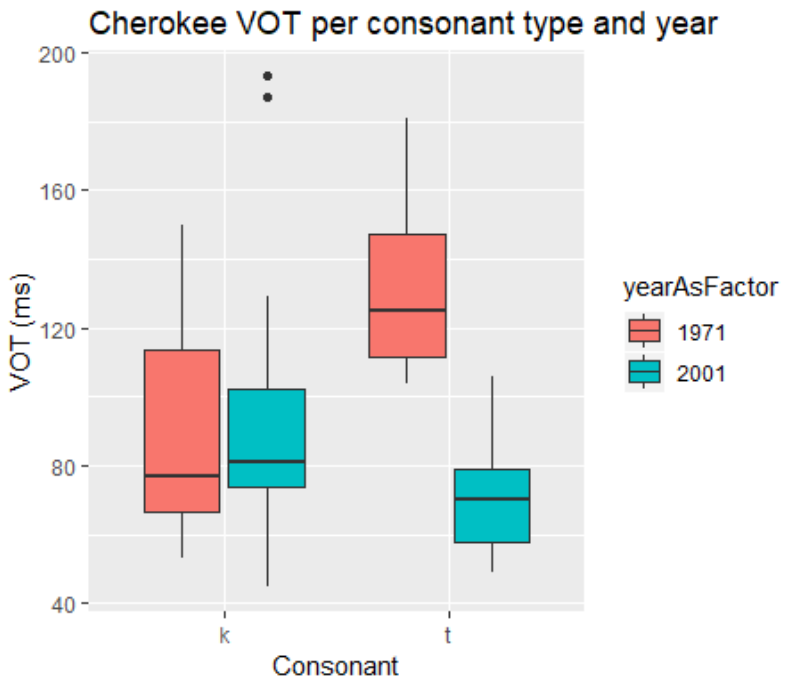
Scatterplot



How to describe the relationship between three variables?

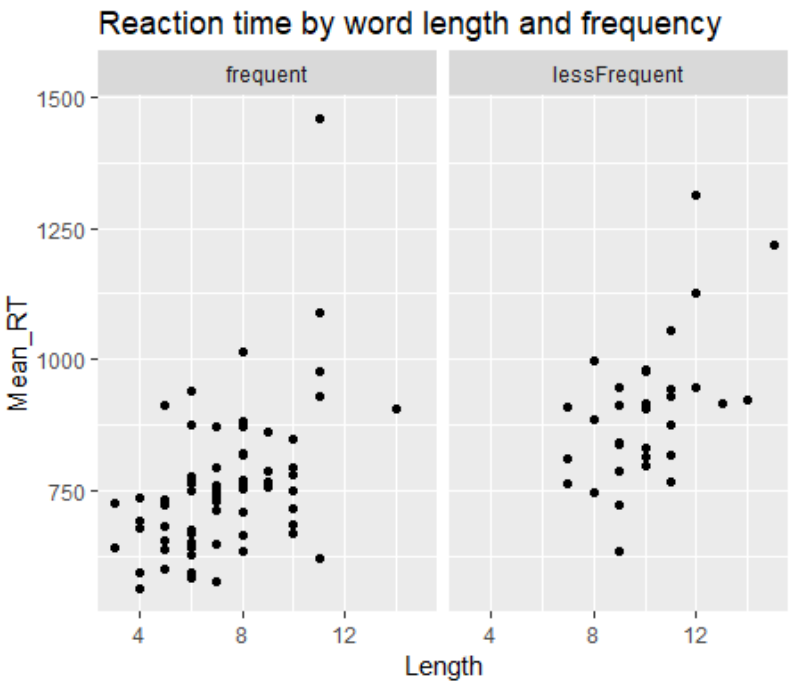
<u>Categorical x categorical x continuous</u>

Side-by-side Boxplot



<u>Categorical x continous x continuous</u>

Scatterplot with facets



Summary

(1) Whenever you build a categorical variable, you are implicitly construction a description of the world, an **ontology**. This might influence your results.

(2) There are numerous types of charts that can be used to understand the distribution of your variables.